

AgRP1 modulates breeding season-dependent feeding behavior in female medaka

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This manuscript provides **fundamental** new insight into the mechanisms linking photoperiod, reproduction function, and feeding activity, using medaka, a genetic model that itself exhibits photoperiodic responses. As well as identifying key neuropeptide genes that are regulated by photoperiod and involved in regulating feeding activity, the authors establish a knockout line for *agrp1* using CRISPR Cas9 - based approach, profiting from the extensive use and development on this methodology in medaka. The combination of the RNAseq and quantitative in situ hybridization analysis with the knockout results as well as the study of ovariectomized fish provides **compelling** evidence implicating *agrp1* in feeding regulation in response to photoperiod and reproductive status.

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Abstract

Feeding and reproduction are known to be closely correlated with each other, and the seasonal breeders show breeding season-dependent feeding behavior. However, most model animals do not have definite breeding seasonality, and the mechanisms for such feeding behavior remain unclear. Here, we focused on female medaka (*Oryzias latipes*); they show breeding season-dependent feeding behavior, and their condition of breeding season can be experimentally controlled by day-length. We first demonstrated that, among previously reported feeding-related peptides (neuropeptides involved in feeding), agouti-related peptide 1 (*agrp1*) and neuropeptide y b (*npyb*) show higher brain expression under the breeding condition than under the non-breeding one. Combined with analysis of *agrp1* knockout medaka, we obtained results to suggest that long day-induced sexually mature condition, especially ovarian estrogenic signals, increase the expressions of *agrp1* in the brain, which results in increased food intake to promote reproduction. Our findings advance the understanding of neural mechanisms of feeding behavior for reproductive success.

Introduction

Feeding behavior is essential to animals for their survival and reproduction and is known to be modulated by various internal and external factors: nutritional status, sexual maturity, temperature, seasonality, etc. This behavior is known to be closely correlated with reproduction (Kauffman & Rissman, 2004 [↗](#)), which is an essential biological activity important for the animal life. Previous studies reported that nutritional-state modulates reproductive behaviors and functions (Chen et al., 2006 [↗](#); Amirjani et al., 2019 [↗](#); Volk et al., 2017 [↗](#); Lynn et al., 2010 [↗](#)). For example, musk shrews show defective sexual behavior under fasted conditions (Temple & Rissman, 2000 [↗](#)). In addition, not a few studies demonstrated that fasting-induced low energy condition suppresses reproduction (Evans & Anderson, 2012 [↗](#); Kalra & Kalra, 1996 [↗](#); Kirkwood et al., 1987 [↗](#); Merry & Holehan, 1979 [↗](#); Hasebe et al., 2016 [↗](#)). Thus, it has been well investigated how nutritional status resulting from feeding modulates reproduction. On the other hand, it has been reported that some animals change their feeding behavior during the breeding season. For instance, the black seabream migrates to the shallow water during the breeding season (Tsuyuki, 2018 [↗](#); Kawai et al., 2020 [↗](#)) where they can get more food, and the white-tailed deer spend more time for feeding under reproductive status (Stone et al., 2017 [↗](#)). Such a close relationship between reproduction and feeding is thought to be important for biological fitness. However, the regulatory mechanisms for breeding season-dependent feeding behavior are still unknown. One possible reason is that most of the model animals appear to have lost the well-defined breeding season. Although the mammalian models, mice and rats, and teleost model zebrafish, have reproductive cycles of about 4-5-days (Nilsson et al., 2015 [↗](#); Peute et al., 1978 [↗](#)), they do not have definite breeding seasonality. Thus, the mechanisms for long-term changes in feeding behavior according to the breeding season have not yet been studied in detail.

Here, as a model animal for the seasonal breeder, we used a teleost fish, medaka (*Oryzias latipes*). Medaka is a useful model animal, whose reproductive status can be experimentally controlled by day length (Robinson & Rugh, 1943 [↗](#); Egami, 1954 [↗](#)) while keeping an appropriate temperature constant. In the long-day (LD) condition (14 h light/10 h dark), female medaka becomes reproductive and regularly spawns every day, while it becomes non-reproductive in the short-day (SD) condition (10 h light/14 h dark). In other words, LD or SD condition can induce breeding or non-breeding season of female medaka, respectively. Thus, medaka enables us to analyze the mechanisms of breeding season-dependent feeding behavior without consideration for possible changes in metabolism and gene expressions due to the changes in ambient temperature.

Regulatory mechanisms of feeding behavior have mainly been analyzed in mammals. These studies reported that some neuropeptides, such as agouti-related peptide (AgRP) and neuropeptide Y (NPY), are involved in the control of feeding and called “feeding-related peptides” as key molecules for the regulation of feeding behavior (Hahn et al., 1998 [↗](#); Aponte et al., 2011 [↗](#); Krashes et al., 2011 [↗](#); Andermann & Lowell, 2017 [↗](#)). Teleosts have also been thought to possess a regulatory mechanism for feeding similar to mammals. In fact, expression of homologous genes coding for feeding-related peptides have been reported (Rønnestad et al., 2017 [↗](#); Conde-Sieira & Soengas, 2016 [↗](#)). On the other hand, although administration of some of them have been suggested to induce feeding behavior in teleosts as well (Rønnestad et al., 2017 [↗](#)), their functions in feeding behaviors still remain unclear.

In the present study, to understand mechanisms of breeding season-dependent feeding behavior, we focused on female medaka, which clearly show seasonal changes in breeding conditions by day length (Mitani et al., 2010 [↗](#); Kanda et al., 2008 [↗](#)) under the fixed appropriate temperature. We first quantified changes in feeding behavior according to the breeding states and found that female medaka under the condition of breeding season (LD) eat more than those under the condition of non-breeding (SD). Therefore, we searched for genes that show breeding state-

dependent changes in expression and found some candidates for feeding-related peptides in medaka. We then analyzed expressions of the candidate genes by using RNA-seq, *in situ* hybridization, and RT-qPCR, and analyzed phenotypes of gene knockout medaka. These results led us to conclude that AgRP1 plays a key role in the breeding season-dependent changes in feeding behavior via ovarian estrogenic signals.

Results

Feeding behavior of female medaka is upregulated in breeding season

To analyze food intake of male and female medaka in breeding/non-breeding seasons, we first established a method for measuring food intake in medaka. In brief, we placed medaka in a white cup, fed brine shrimp to medaka in all-you-can-eat style for 10 min, and counted the leftover brine shrimp in the cup with a “shrimp-counter” system (called Japanese “Wanko-soba” like method, Figure 1-Source Code 1 and Figure 1-Supplementary Figure 1). We used this system to analyze food intake of male and female medaka under the breeding condition equivalent to that in breeding season (kept under long day (LD) condition) or under the non-breeding condition equivalent to that in non-breeding season (short day (SD) condition) (Figure 1A [↗](#); Kanda et al., 2008 [↗](#)). We found that female medaka under the breeding (LD) condition eat more than those under the non-breeding (SD) condition (Figure 1B [↗](#); $p = 0.02519$). In contrast to female, in males there was no significant difference in food intake between the breeding and non-breeding condition (Figure 1C [↗](#); $p = 0.6540$). Since these results demonstrated that females, not males, show breeding season-dependent feeding behavior, we focused only on female medaka in the following analyses on neuronal mechanism for breeding season-dependent feeding behavior. Next, to examine which gene products modulate feeding behavior of female medaka in breeding season, we performed mRNA-sequencing (RNA-seq) using the whole brain of female medaka in breeding condition (LD) and non-breeding condition (SD) (Figure 1-Supplementary Figure 2A). Overall, 1025 genes showed significantly different expression between LD and SD female medaka. Figure 1-Supplementary Figure 2B shows a heat map of representative genes mainly related to neuroendocrine system, which were differently expressed between LD and SD females. Among the conventional candidate feeding-related neuropeptides, we identified two kinds of neuropeptides, *agrp1* and *npyb*, both of which showed higher expression in LD than in SD (Figure 1-Supplementary Figure 2C and 2D). Both AgRP and NPY are known to have orexigenic effects mainly in mice (Schwartz et al., 2000 [↗](#); Andermann & Lowell, 2017 [↗](#)). Therefore, in the subsequent analyses, we focused on *agrp1* and *npyb* as candidate genes that modulate breeding season-dependent feeding behavior in female medaka.

AgRP1, NPYa and NPYb may be the “feeding-related peptides” in female medaka

Medaka has two *agrp* paralogues, *agrp1* and *agrp2* and two *npy* paralogues, *npya* and *npyb*, which arose from third round whole genome duplication early in the teleost lineage (Liu et al., 2019 [↗](#); Sundstrom et al., 2008 [↗](#)). Therefore, we next examined the anatomical distribution of neurons expressing *agrp1*, *npyb*, and their paralogs in the female brain by *in situ* hybridization (ISH). We found that *agrp1*- and *npyb*-expressing neurons are distributed in local brain regions (Figures 2A [↗](#), 2B [↗](#) and 2E [↗](#)), while *npya*- and *agrp2*-expressing neurons are more widely distributed (Figures 2A [↗](#), 2C [↗](#) and 2D [↗](#)) in the brain. The *agrp1* neurons were distributed in the nucleus ventralis tuberis (NVT) of the hypothalamus (Figure 2B [↗](#)), while *agrp2* neurons were expressed in the telencephalon and in the hypothalamus (Figure 2C [↗](#)). On the other hand, *npya* neurons

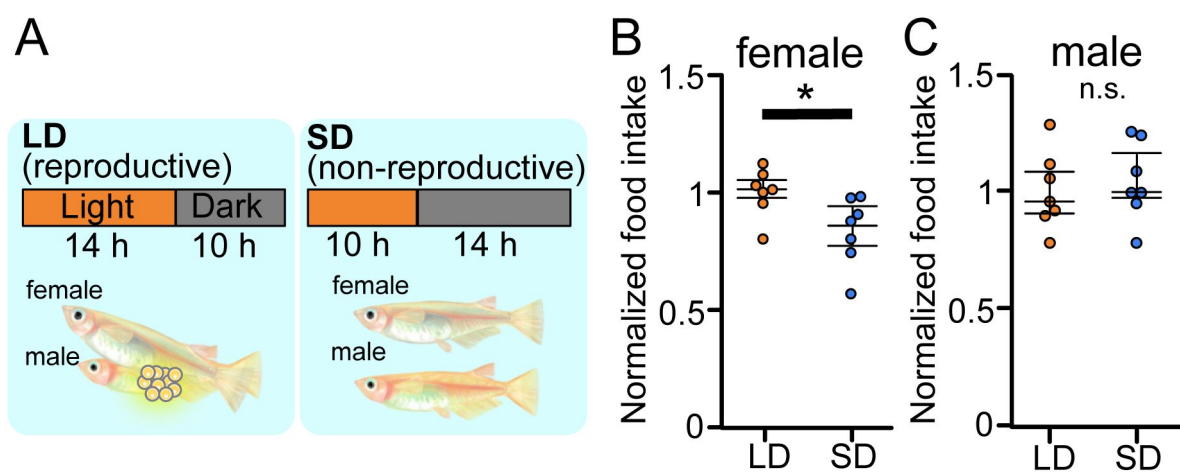


Figure 1

Reproductive female medaka show larger amount of food intake.

(A) Light conditions for breeding and non-breeding status. (B) Food intake (10 mins) of female medaka in LD (orange; $n = 7$) and SD (blue; $n = 7$) conditions, normalized to the amount of artemia eaten by medaka in LD (breeding) condition ($p = 0.02519$, $U = 42.5$). (C) Food intake (10 mins) of male medaka in LD (orange; $n = 7$) and SD (blue; $n = 7$) conditions, normalized to the amount of artemia eaten by medaka in LD (breeding) condition ($p = 0.6540$, $U = 20.5$). Mann-Whitney U test, * $p < 0.05$. n.s., not significant.

were distributed more widely from telencephalon to hypothalamus (**Figure 2D**). *npyb* neurons were distributed locally in the nucleus ventralis telencephali pars dorsalis (Vd) of the telencephalon (**Figure 2E**).

In mice, *agrp* is known to be only expressed in the hypothalamus and mostly co-expressed with *npy* (Hahn et al., 1998), and these AgRP/NPY neurons are known to regulate mammal feeding behavior (Shutter et al., 1997; Broberger et al., 1998; Ollmann et al., 1997; Takahashi & Cone, 2005). In medaka, on the other hand, *agrp1* signals were not observed in *npya* neurons (**Figure 2F** left), although the both genes were expressed in the hypothalamus. In addition, *agrp2* signals were not observed in hypothalamic *npya* neurons (**Figure 2F** right), either. These results suggest that AgRP and NPY are not co-expressed in medaka. Since AgRP and NPY of medaka showed different expressing patterns compared with other animals such as mice, we examined whether they act as modulators of feeding. We divided female medaka in LD condition into two groups; one group was kept under normally fed condition (Fed), and the other was kept under 2-week food restricted condition (FR). We then analyzed whole-brain expressions of these four genes. RT-qPCR analysis demonstrated that 2-week FR increased the expression of *agrp1* (**Figure 2G**; $p = 0.001243$) but decreased that of *npyb* (**Figure 2J**; $p = 0.0003108$), suggesting that the two peptides are involved in feeding in an opposite manner. On the other hand, *agrp2* did not significantly change their expressions between Fed and FR conditions (**Figure 2H**; $p = 0.9551$). Although *npya* did not significantly change their expressions between Fed and FR conditions (**Figure 2I**; $p = 0.2319$), it may be possible that *npya* expression changed in a specific brain region, since *npya* neurons are widely distributed in various brain regions as described above (**Figure 2D**). Since it is suggested that NPY released from hypothalamic *npy*-expressing neurons controls food intake in mice (Kohno & Yada, 2012), we also examined the *npya*-expression in medaka hypothalamus by ISH. We counted *npya*-expressing neurons in each hypothalamic region and compared them between Fed and FR female medaka (Figure 2-Supplementary Figure 1). We found that the total number of *npya*-expressing neurons in the hypothalamus was significantly larger in Fed compared with FR (Figure 2-Supplementary Figure 1A; $p = 0.02857$). Here, significant increase in cell number was observed in NRL ($p = 0.02857$) and NAT ($p = 0.02857$), but not in NVT ($p = 0.1143$) (Figure 2-Supplementary Figure 1B-D). On the other hand, the expression of *agrp2* did not show remarkable difference in the hypothalamus under food restriction or not (Figure 2-Supplementary Figure 1E). Thus, the results suggest that AgRP1, NPYa and NPYb may be the “feeding-related peptides” in female medaka.

Both *agrp1* and *npyb* show higher expression levels in LD than in SD female medaka

To further examine the result of RNA-sequencing (Figure 1-Supplementary Figure 2), we compared expression of *agrp1* and *npyb* between the female medaka under the breeding condition (LD) and those under the non-breeding condition (SD), using ISH and whole-brain RT-qPCR (**Figure 3**). First, we performed whole-brain RT-qPCR and found that the expression level of *agrp1* was higher in LD than in SD female (**Figure 3A**, $p = 0.001865$). In ISH experiments, we observed larger number of *agrp1*-expressing neurons in LD than in SD females (**Figure 3B** and **3C** ($p = 0.02828$), Figure 3-Supplementary Figure 1). Since the expression level of *agrp1* was higher in LD than that in SD (**Figure 3A**), higher expression of *agrp1* under the breeding condition may be due to the increase in the number of neurons expressing *agrp1*. On the other hand, *npyb* expression in RT-qPCR was significantly higher in LD than that in SD (**Figure 3D**; $p = 0.0001554$), although ISH analysis demonstrated that *npyb*-expressing cell number was not significantly different between LD and SD (**Figure 3E**; $p = 0.4206$). These results suggest that the expression level for each neuron increased in LD compared with SD. Thus, higher expression of *npyb* under the breeding condition may be due to the increase of expressions in each neuron expressing *npyb*.

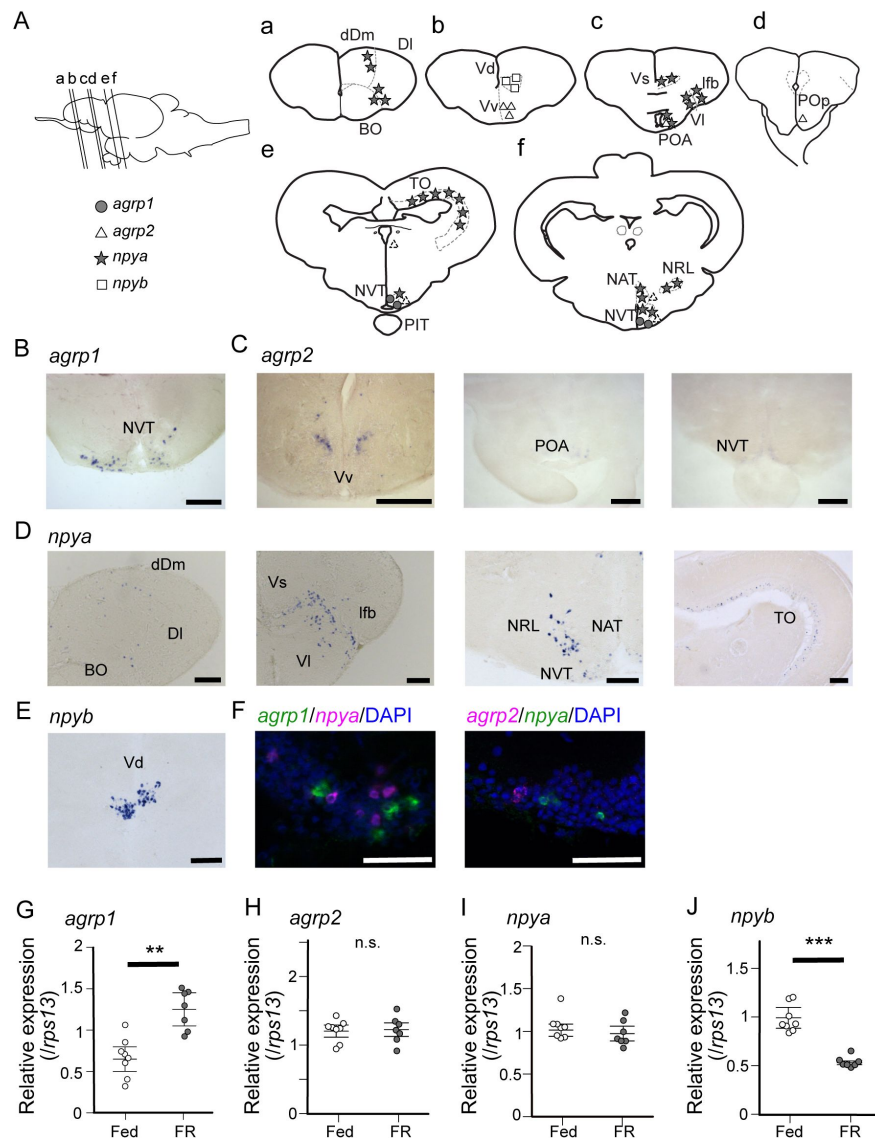


Figure 2

***agrp1*- and *npyb*-expressing neurons are distributed locally, while *npya*- and *agrp2*- expressing neurons are distributed more widely in the brain.**

(A) Illustration of the lateral view of medaka brain and distributions of cell bodies expressing each *agrp* or *npy* gene. The oblique lines labeled with a-e indicate the level of the frontal sections in a-e. (a-e) Illustrations of frontal sections showing the distribution of neurons expressing each gene. The localization of neurons is indicated in the right half of the illustrations. BO bulbus olfactorius, dDm dorsal region of area dorsalis telencephali pars medialis, DI area dorsalis telencephali pars lateralis, lfb lateral forebrain bundle, NAT nucleus anterior tuberis, NRL nucleus recessus lateralis, NVT nucleus ventralis tuberis, PIT pituitary, POA area preoptica, POp nucleus preopticus pars paravocellularis, TO tectum opticum, Vd area ventralis telencephali pars dorsalis, VI area ventralis telencephali pars lateralis, Vs area ventralis telencephali pars supra commissuralis, Vv area ventralis telencephali pars ventralis. *agrp1* (gray circle), *agrp2* (open triangle: high expression, dotted triangle: low expression), *npya* (star), *npyb* (open square). (B) *agrp1*-expressing neurons are localized in NVT. Scale bar: 100 μ m. (C) *agrp2*-expressing neurons are observed in Vv, POA, and NVT. Scale bar: 100 μ m. (D) *npya*-expressing neurons are distributed in dDm, DI, BO, Vs, VI, lfb, NRL, NAT, NVT, and TO. Scale bar: 100 μ m. (E) *npyb*-expressing neurons are localized in Vd. Scale bar: 100 μ m. (F) Left: *agrp1* (green) and *npya* (magenta) are distributed in NVT, but the two genes are not co-expressed. Right: *agrp2* (magenta) and *npya* (green) are distributed in NVT, but the two genes are not co-expressed. Scale bars: 50 μ m. (G-J) *agrp* and *npy* expressions in the whole brain of female medaka under normally fed condition (Fed; white; n = 7) or 2-week food restricted (FR; gray; n = 8). (G) *agrp1* ($p = 0.001243$, $U = 2$) (H) *agrp2* ($p = 0.9551$, $U = 29$) (I) *npya* ($p = 0.2319$, $U = 39$) (J) *npyb* ($p = 0.0003108$, $U = 56$) Mann-Whitney U test, ** $p < 0.01$, *** $p < 0.001$. n.s., not significant.

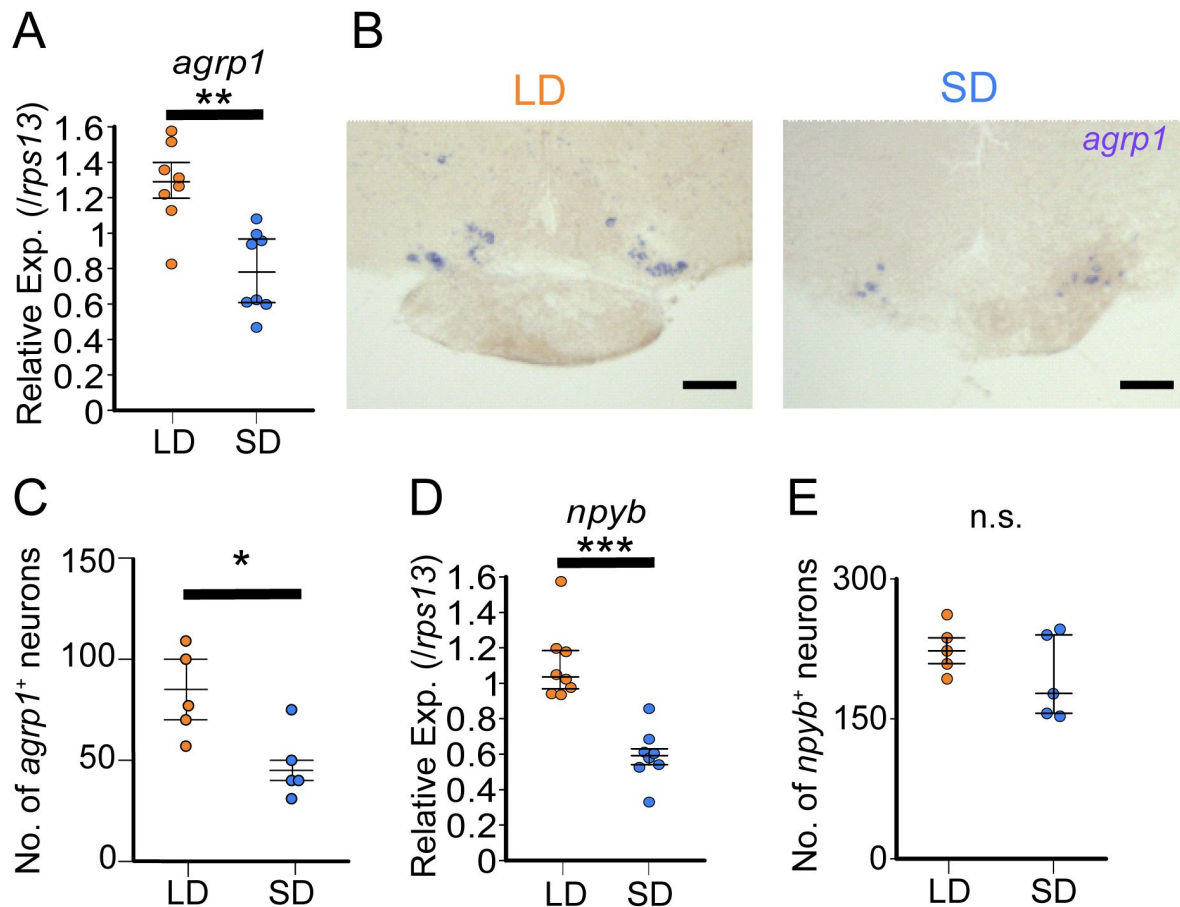


Figure 3

***agrp1* and *npyb* show higher expression levels in LD than in SD female.**

(A) *agrp1* expression in the whole brain of LD (orange; n = 8) and SD (blue; n = 8) female medaka. ($p = 0.001865$, $U = 60$) (B) ISH of *agrp1*-expressing neurons in LD and SD female medaka. Scale bar: 100 μ m. (C) The number of neurons expressing *agrp1* in LD (orange; n = 5) and SD (blue; n = 5). ($p = 0.02828$, $U = 23$) (D) *npyb* expression in the whole brain of LD (orange; n = 8) and SD (blue; n = 8) female medaka. ($p = 0.0001554$, $U = 64$) (E) The number of neurons expressing *npyb* in LD (orange; n = 5) and SD (blue; n = 5). ($p = 0.4206$, $U = 17$) The upper, middle, and lower bars show the third quartile, median, and the first quartile, respectively. Mann-Whitney U test, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. n.s., not significant.

In juvenile female medaka, expression levels of neither *agrp1* nor *npyb* shows significant change according to the day-length

The results thus far indicates that expressions of *agrp1* and *npyb* are upregulated in female medaka under the condition of breeding season. Since the breeding/non-breeding state is experimentally controlled by day-length (LD/SD) in the present study, we examined which of the two factors, day-length itself or substance(s) from LD-induced mature ovary, regulates the expression of *agrp1* and *npyb*. Here, we used sexually immature juvenile medaka and compared their whole-brain expressions of *agrp1* and *npyb* under LD/SD conditions using RT-qPCR (**Figure 4**). We found that expression levels of neither *agrp1* nor *npyb* show significant difference between LD and SD (**Figure 4A** ($p = 0.4179$) and **4B** ($p = 0.3357$)). Furthermore, food intake of juvenile female was not different between LD and SD (**Figure 4-Supplementary Figure 1**; $p = 0.7197$). These results suggest that neither of them is regulated directly by day length itself. Instead, the gene expression is suggested to be regulated by LD-induced sexual maturity.

Estrogen, which is released from mature ovary, may affect the expression of *agrp1*

Among various factors associated with ovarian maturity, estrogens are known to be abundantly released from mature ovary and play important roles in reproductive readiness, sexual behavior, and so on (Jennings & de Lecea, 2020; Naftolin et al., 2007; Adachi et al., 2007; Clarkson & Herbison, 2009; Wintermantel et al., 2006; Micevych & Meisel, 2017; Melo & Ramsdell, 2001). Among the ovarian estrogens, 17 β -estradiol (E2) is the major factor important for reproduction (Kanda et al., 2011; Kelly & Qiu, 2010), and the blood E2 concentration of LD-conditioned female medaka is higher than those of SD (Ikegami et al., 2022). Thus, we hypothesized that E2 regulates the expression of *agrp1* and *npyb* under the condition of breeding season. We analyzed the expression of *agrp1* and *npyb* in sham-operated (Sham), ovariectomized (OVX, fish with surgical ablation of the ovary), and OVX medaka with E2-administration (OVX+E) (**Figure 5A–5B**). The OVX medaka were allowed to survive at least for two weeks to clear the endogenous E2 (Kayo et al., 2020), and spawning of all the Sham medaka were confirmed for three consecutive days. By using whole-brain RT-qPCR, we found that OVX induces significantly lowered *agrp1* expression than Sham, and OVX+E shows a tendency to recover *agrp1* expression compared with OVX (**Figure 5A**; Sham vs OVX: $p = 0.04310$, OVX vs OVX+E: $p = 0.05790$, Sham vs OVX+E: $p = 0.2000$), which suggests that the ovarian E2 regulates *agrp1* expression. On the other hand, the expression levels of *npyb* did not show significant differences among the three groups (**Figure 5B**; Sham vs OVX: $p = 0.1386$, OVX vs OVX+E: $p = 0.9991$, Sham vs OVX+E: $p = 0.08120$). In addition, food intake of OVX female was not significantly different between LD and SD (**Figure 5-Supplementary Figure 1**; $p = 0.7308$), which suggests that ovarian signal may be important for breeding season-dependent feeding behavior. Therefore, we focused more on the estrogenic regulation of *agrp1* expression.

Estrogens act mainly by interacting with estrogen receptors (Chen et al., 2022). Medaka has three kinds of estrogen receptors; *esr1*, *esr2a* and *esr2b* (Tohyama et al., 2016; Kinoshita et al., 2009), and all of them have been reported to be expressed in NVT (Zempo et al., 2013), in which *agrp1* was also expressed (**Figure 2B**). Therefore, we examined co-expression of these *esr* genes and *agrp1*. As shown in **Figure 5C** and **5D**, *esr2a* signal was clearly co-expressed in some *agrp1*-expressing neurons, which strongly suggests that E2 affects the expression of *agrp1* via *esr2a* in those neurons of NVT.

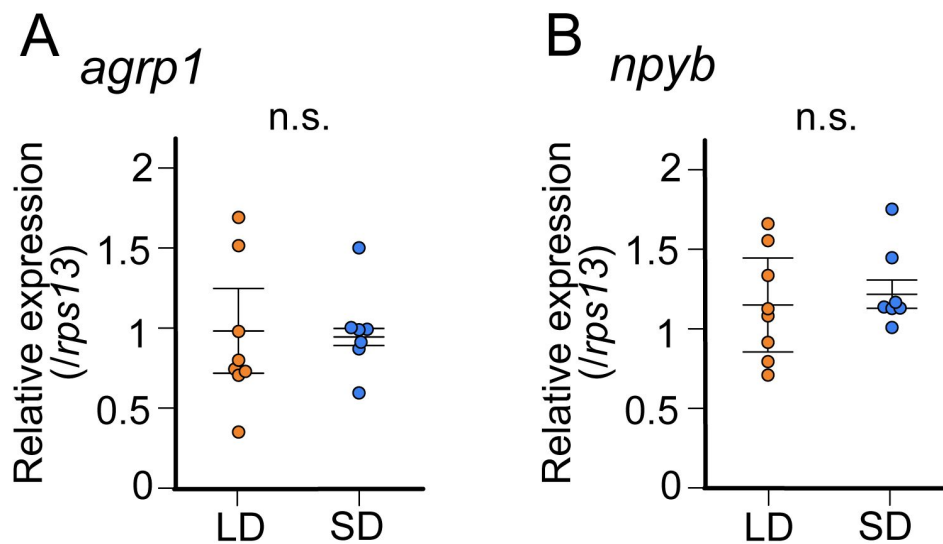


Figure 4

In juvenile female medaka, expression levels of neither *agrp1* nor *npyb* shows significant change according to the day-length.

(A) *agrp1* expression in the brain of juvenile female medaka. ($p = 0.4179$, $U = 21$) (B) *npyb* expression in the brain of juvenile female medaka. ($p = 0.3357$, $U = 19$) LD: orange, $n = 8$; SD: blue, $n = 7$. The upper, middle, and lower bars show the third quartile, median, and the first quartile, respectively. Mann-Whitney U test. n.s., not significant.

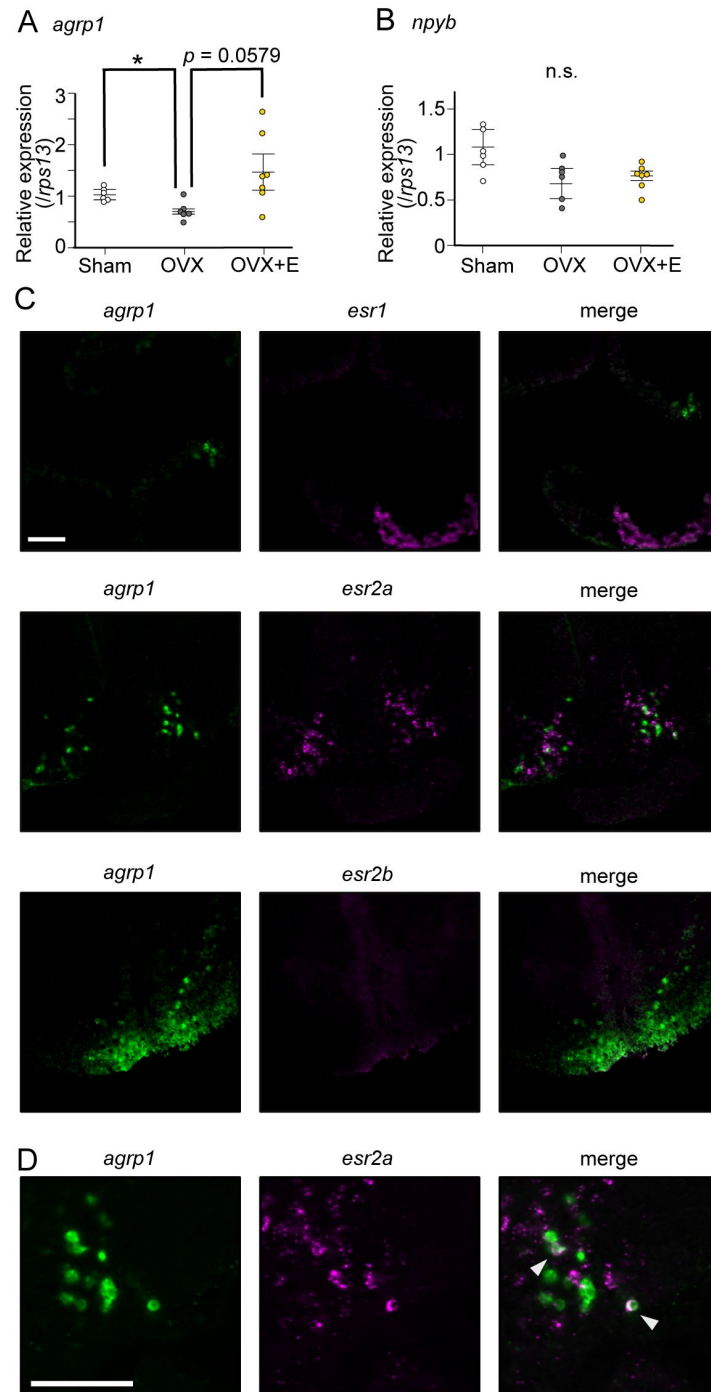


Figure 5

Estrogen, which is secreted from mature ovary, may affect the expression of *agrp1*.

(A) Relative expression of *agrp1* in Sham (white; $n = 6$), OVX (Ovariectomized medaka, gray; $n = 6$), and OVX + E (OVX medaka kept in the tank containing E2, yellow; $n = 7$). Sham vs OVX: $p = 0.04310$, OVX vs OVX+E: $p = 0.05790$, Sham vs OVX+E: $p = 0.2000$. (B) Relative expression of *npyb* in Sham, OVX, and OVX + E. Sham vs OVX: $p = 0.1386$, OVX vs OVX+E: $p = 0.9991$, Sham vs OVX+E: $p = 0.08120$. The upper, middle, and lower bars show the third quartile, median, and the first quartile, respectively. Steel-Dwass test, * $p < 0.05$. n.s., not significant. (C) Photographs of brain slices after experiments of double *in situ* hybridization (*agrp1* (green) and *estrogen receptors* (magenta)). (D) Expanded photograph of *agrp1* and *esr2a* co-expressing neurons (white arrowhead). Scale bars: 50 μm .

***agrp1*^{-/-} female medaka show a decrease in food intake and in the number of fertilized eggs**

Our present experimental evidence thus far highlights the importance of *agrp1* as the factor modulating the season-dependent feeding behavior in medaka. To analyze the function of AgRP1 in medaka, we generated knockout medaka of *agrp1* (*agrp1*^{-/-}) by using CRISPR/Cas9. The designed CRISPR guide RNA cleaved targeted sites of *agrp1* (exon3, Figure 6-Supplementary Figure 1A), and we obtained *agrp1*^{-/-} medaka, which has lots of amino acid changes in functional site for AgRP1 (Figure 6-Supplementary Figure 1B). In *agrp1*^{-/-} brain, AgRP1 immunoreactive signals, which were observed in WT, were not found (Figure 6-Supplementary Figure 1C). These suggested that *agrp1*^{-/-} possess nonfunctional AgRP. As for phenotype of the knockout, the *agrp1*^{-/-} female medaka appeared skinny and the body weight was significantly lower than that of *agrp1*^{+/+} (Figure 6A; body weight: $p = 0.04113$; abdominal length: $p = 0.002165$). In addition, abdominal height of *agrp1*^{-/-} was also smaller than that of *agrp1*^{+/+}, while the body length was not significantly different (Figure 6A; body length: $p = 0.3939$). We next analyzed food intake of *agrp1*^{-/-} female medaka in LD condition (breeding). As shown in Figure 6B ($p = 0.004329$) and Figure 6-Supplementary Figure 2 (*agrp1*^{+/+} vs *agrp1*^{-/-}: $p = 0.5470$, *agrp1*^{+/+} vs *agrp1*^{-/-}: $p = 0.009353$, *agrp1*^{-/-} vs *agrp1*^{-/-}: $p = 0.01234$), we found that LD *agrp1*^{-/-} female medaka eat less than *agrp1*^{+/+}. Then, we kept *agrp1*^{-/-} medaka in LD or SD condition and compared their food intake. In contrast to *agrp1*^{+/+}, *agrp1*^{-/-} in LD condition did not show a significant increase in food intake compared with SD (Figure 6C; $p = 0.5953$). We also examined whether loss of AgRP1 affects reproductive function. Whereas the *agrp1*^{-/-} females were fertile, the pairs of *agrp1*^{-/-} female and *agrp1*^{+/+} male resulted in fewer spawned eggs than *agrp1*^{+/+} pairs (Figure 6D; $p = 0.008658$). In addition, the ovarian size of *agrp1*^{-/-} appeared to be smaller than *agrp1*^{+/+} (Figure 6E left). In particular, since relative ovarian weight normalized by body weight (Gonadosomatic index: GSI) of *agrp1*^{-/-} female tended to be marginally smaller than *agrp1*^{+/+} (Figure 6E right; $p = 0.06494$), the smaller body size of *agrp1*^{-/-} (Figure 6A) may drastically affect ovarian morphology. Since the number of spawned eggs was decreased in *agrp1*^{-/-} female, we analyzed gene expressions of gonadotropins which should affect ovarian maturation. Oocyte maturation and ovulation are known to be regulated by gonadotropins, follicular-stimulating hormone (FSH) and luteinizing hormone (LH). As shown in Figure 6F, *agrp1*^{-/-} females showed lower levels of expression of gonadotropin genes (*fshb* and *lhb*; *lhb*: $p = 0.008658$; *fshb*: $p = 0.02597$), which suggests that loss of function of *agrp1* impaired breeding season-dependent feeding behavior and led to attenuation of reproductive functions, especially the decrease in number of spawned eggs in the breeding season.

Discussion

In the present study, we took advantage of female medaka which clearly shows breeding season-dependent feeding behavior and found that neuropeptides, *agrp1* and *npyb*, show higher expression under the breeding condition than under the non-breeding condition. We also obtained results to suggest that the expression of both *agrp1* and *npyb* changes depending on nutritional status of female medaka. In addition, ovariectomy and E2 administration changed expression of *agrp1* but not *npyb*, suggesting that increased release of ovarian E2 in breeding season upregulates the *agrp1* expression, which results in facilitation of female feeding behavior. Finally, loss-of-function mutation of AgRP1 decreased the amount of food intake and the number of spawned eggs. The present results suggest that breeding season-dependent feeding behavior can be modulated by the increased expression of *agrp1* upregulated by increased release of ovarian estrogen in the breeding season (Figure 7). To date, not a few previous reports have shown the influence of nutritional status on reproduction (Evans & Anderson, 2012; Kalra & Kalra, 1996; Kirkwood et al., 1987; Merry & Holehan, 1979; Hasebe et al., 2016). On the other hand, although seasonal breeders have been reported to show changes in feeding behavior during the

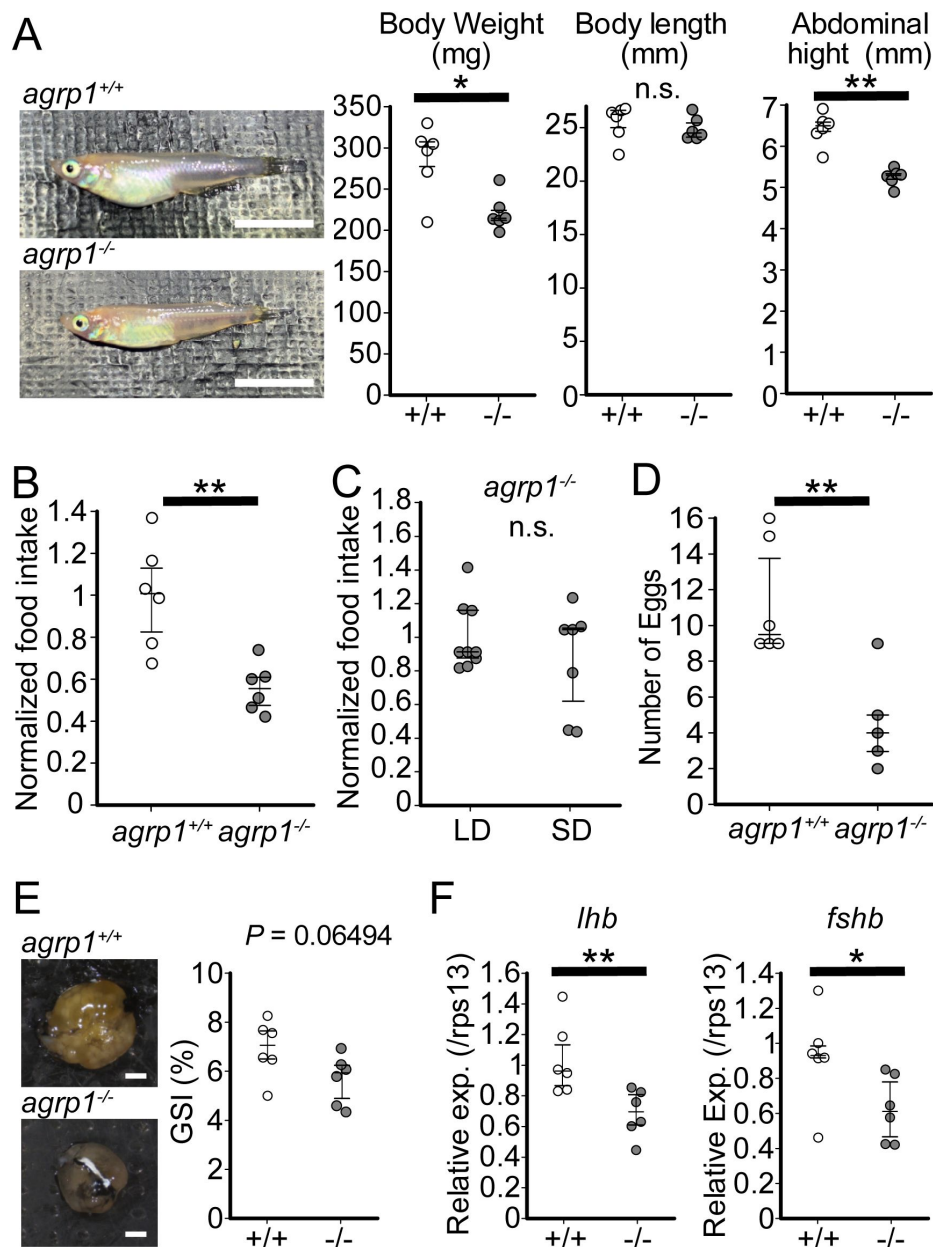


Figure 6

***agrp1*^{-/-} female medaka show decrease in food intake and the number of fertilized eggs.**

(A) Lateral views of representative *agrp1*^{+/+} and *agrp1*^{-/-} female medaka (Left), and body weight, body length, and abdominal height (right) of *agrp1*^{+/+} (white; n = 6) and *agrp1*^{-/-} (gray; n = 6). All the fish are 4-month-old adult medaka. Scale bar: 1 cm. body weight: $p = 0.04113$, $U = 31$; body length: $p = 0.3939$, $U = 24$; abdominal length: $p = 0.002165$, $U = 36$. (B) Food intake (10 min) of *agrp1*^{+/+} (white; n = 6) and *agrp1*^{-/-} (gray; n = 6) female medaka. Each amount of food intake is normalized by the average number of that of *agrp1*^{+/+} medaka. ($p = 0.004329$, $U = 35$) (C) Food intake (10 min) of LD *agrp1*^{-/-} (white; n = 9) and SD *agrp1*^{-/-} (gray; n = 7) female medaka. Each amount of food intake is normalized by the average number of that of LD *agrp1*^{-/-} medaka. ($p = 0.5953$, $U = 37$) (D) The number of eggs spawned by *agrp1*^{+/+} (white; n = 6) and *agrp1*^{-/-} (gray; n = 5) female medaka in a day. Each female was paired with a wildtype male. ($p = 0.008658$, $U = 28.5$). (E) Photograph of ovary in representative *agrp1*^{+/+} and *agrp1*^{-/-} female (left) and the gonado-somatic index (GSI, right). n = 6 of each group. ($p = 0.06494$, $U = 30$) Scale bar: 1 mm. (F) Expression of gonadotropin genes (*lhb* and *fshb*) in the pituitary of *agrp1*^{+/+} (white; n = 6) and *agrp1*^{-/-} (gray; n = 6) female medaka. *lhb*: ($p = 0.008658$, $U = 34$); *fshb*: ($p = 0.02597$, $U = 32$). The upper, middle, and lower bars show the third quartile, median, and the first quartile, respectively. Mann-Whitney U test, * $p < 0.05$, ** $p < 0.01$. n.s., not significant.

breeding season (Tsuyuki, 2018 [↗](#); Kawai et al., 2020 [↗](#)), its neuroendocrine mechanism have largely remained enigmatic. Our present results may provide a neuroendocrinological model for the mechanisms that play a key role in the control of breeding season-dependent feeding behavior in teleosts.

Feeding-related peptides AgRP and NPY in medaka

Here, we demonstrated that female medaka eat more under the condition of breeding season (Figure 1B [↗](#)). Various kinds of neuropeptides in the brain have been suggested to control feeding, and these are generally called “feeding-related peptides” (Funahashi et al., 2003 [↗](#)). In the present study, we first used a seasonally breeding model teleost medaka and searched for the “feeding-related peptides” involved in seasonal feeding behavior. A whole-brain RNA-seq analysis using female medaka under the breeding condition (LD) and non-breeding condition (SD) revealed two kinds of feeding-related peptides, *agrp1* and *npyb*, which show different expression levels between LD and SD (Figure 1-Supplementary Figure 2 [↗](#)). In mammals, AgRP and NPY are known to have orexigenic function and are co-expressed in hypothalamic neurons (Schwartz et al., 2000 [↗](#); Hahn et al., 1998 [↗](#)). Previous studies in mammals (Schwartz et al., 2000 [↗](#); Andermann & Lowell, 2017 [↗](#); Muroi & Ishii, 2016 [↗](#)) have suggested neural mechanisms of appetite including functions of AgRP and NPY. However, such mechanisms in non-mammalian vertebrates such as teleosts (Rønnestad et al., 2017 [↗](#); Blanco & Soengas, 2021 [↗](#)) have not yet been clarified. Our present study using medaka has shown possible functions of AgRP and NPY in teleost feeding behavior, especially in a breeding season-dependent manner.

Our present study using medaka showed that female medaka express *agrp1* in hypothalamus, and food restriction increases the *agrp1* expression (Figure 2B [↗](#) and 2G [↗](#)). It has been reported that leptin receptor-knockout medaka show higher food intake and higher expressions of *agrp1* and *npya* than wild type, whereas the expression of *agrp2* and *npyb* remained to be analyzed (Chisada et al., 2014 [↗](#)). Zebrafish has also been used as a model animal in teleosts. In zebrafish, food restriction increased *agrp1* (Song et al., 2003 [↗](#); Opazo et al., 2018 [↗](#)) expression, and transgenic overexpression of *agrp1* caused gain of body weight (Song & Cone, 2007 [↗](#)), as in mammals (Graham et al., 1997 [↗](#); Adam et al., 2002 [↗](#); Hahn et al., 1998 [↗](#); Ilnytska & Argyropoulos, 2008 [↗](#)). It has also been reported that *agrp1* knockout zebrafish eat less than the wild type (Shainer et al., 2019 [↗](#)), although loss of AgRP in mice showed little effect on food intake (Qian et al., 2002 [↗](#)). Our present results and these previous studies strongly support that *agrp1* regulates feeding and may act as an orexigenic factor in teleosts. On the other hand, *agrp2* neurons were distributed mainly in telencephalon, and its weak expressions were also observed in the POA and the hypothalamus (Figure 2C [↗](#)), which is different from the results in zebrafish (Shainer et al., 2017 [↗](#)). In the present study, food restriction did not remarkably affect the *agrp2* expression in medaka (Figure 2 [↗](#) and Figure 2-Supplementary Figure 1E), and AgRP2 in zebrafish is suggested to play an important role in stress response, not feeding (Shainer et al., 2019 [↗](#)). Thus, it is highly probable that *agrp2* is involved in functions other than feeding in female medaka.

Furthermore, we demonstrated that *npya* is expressed in multiple brain regions including hypothalamus in medaka (Figure 2D [↗](#)), which is similar to mammals (Gray & Morley, 1986 [↗](#)) and zebrafish (Yokobori et al., 2012 [↗](#); Jeong et al., 2018 [↗](#)). We also showed that *npya* is not co-expressed with *agrp1* nor *agrp2* (Figure 2F [↗](#)) as in the zebrafish (Jeong et al., 2018 [↗](#)), suggesting that The relationship between NPY and AgRP of teleosts may be different from that of mammals, in which most of the *agrp*-expressing neurons co-express *npy* (Hahn et al., 1998 [↗](#)). Moreover, our present study also showed that *npyb* expression is localized in telencephalon (Figure 2E [↗](#)), which is similar to the previous report using tiger puffer (Kamijo et al., 2011 [↗](#)). Previous studies of NPY in zebrafish showed that zebrafish has only one type of NPY (NPYa) (Söderberg et al., 2000 [↗](#); Larsson et al., 2009 [↗](#)) and has lost NPYb during evolution. Like in mammals (Marks et al., 1992 [↗](#); Clark et al., 1984 [↗](#); Glenn Stanley et al., 1986 [↗](#); Marks & Waite, 1997 [↗](#); Baldock et al., 2009 [↗](#)), food restriction in zebrafish increased the *npya* expression in the hypothalamus (Song et al.,

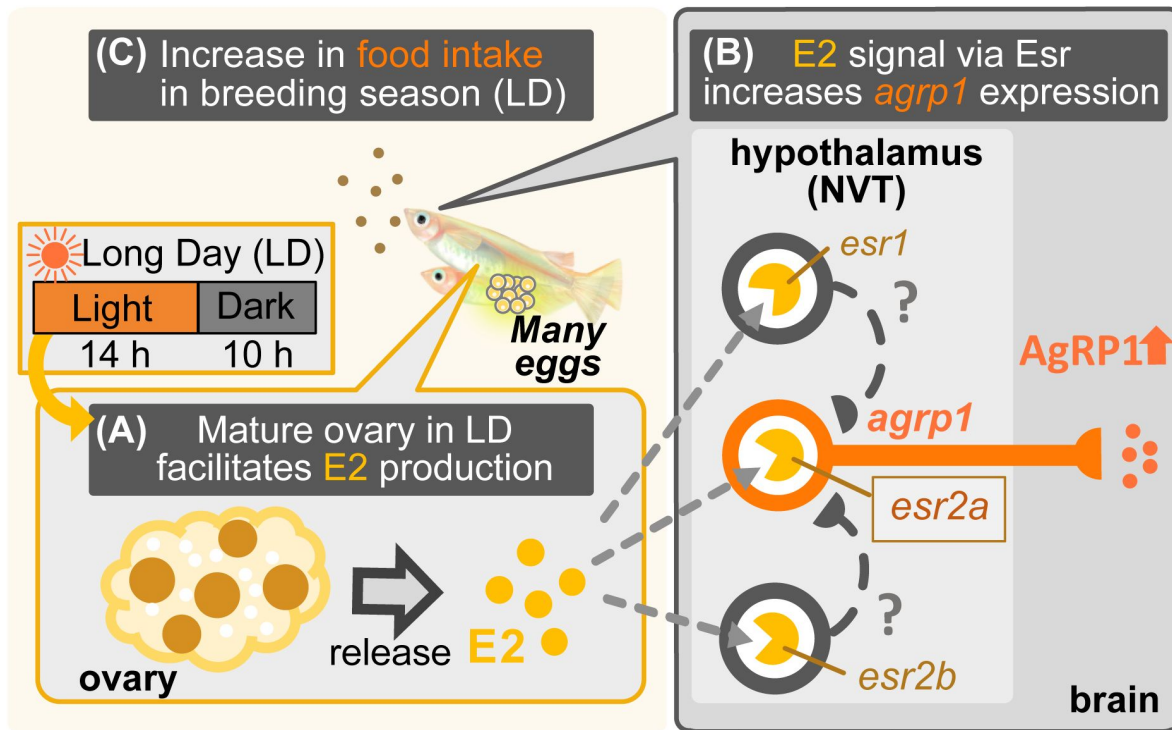


Figure 7

Illustration of mechanisms of breeding season-dependent feeding behavior in medaka suggested by the present study

(A) Long day (LD) condition in breeding season induces ovarian maturation, which facilitates release of estrogen (E2) from the mature ovary. (B) High concentration of serum E2 increases *agrp1* expression in the brain via the estrogen receptors, especially, *esr2a*. (C) Higher expression of AgRP1 is suggested to activate neural circuitry for feeding, which leads to an increase in food intake and egg spawning of female medaka in the breeding season.

2003 [Opazo et al., 2018](#)), and intracerebroventricular administration of NPYa increased food intake (Yokobori et al., 2012). Although these zebrafish studies suggest that NPYa may increase food intake, it is still debatable, since body weight was not significantly different between *npya* knockout zebrafish and wild type (Shiozaki et al., 2020). Interestingly, by analyzing both *npya* and *npyb* expression in medaka of different nutritional conditions, we found that food restriction decreased the *npya*-expressing cell number in the hypothalamus (Figure 2-Supplementary Figure 1) and *npyb* expression level (Figure 2J). These changes in *npya* and *npyb* expressions are not consistent with previous studies using other conventional model animals described above (Yokobori et al., 2012; Marks et al., 1992; Clark et al., 1984; Glenn Stanley et al., 1986; Marks & Waite, 1997; Baldock et al., 2009). Our present study may suggest that the function of *npy* may be different among teleosts. In addition, *npyb* expression was increased under the breeding condition (LD), while LD female showed increase in food intake (Figure 3F). Thus, decreased expression of *npyb* by food restriction (Figure 2J) may suggest that the change in *npyb* expression reflects nutritional condition in medaka. Thus, future study of *npya* and *npyb* functions in the control of feeding will be necessary.

As described above, we found that *agrp1*, *npya*, and *npyb* change expression levels in response to nutritional status. Among these three genes, we suggest that *agrp1* most probably affects relatively long-term feeding in the breeding season, which agrees well with the recent studies in mice showing the function of AgRP as a long-term orexigenic factor. In mice, it has been reported that intracerebroventricular administration of AgRP increases food intake for one week (Hagan et al., 2000), and stimulation of receptors expressed in AgRP neurons triggers AgRP release, leading to an increase in food intake for three days (Nakajima et al., 2016). Our present study also suggests a long-term (seasonal) orexigenic effect of AgRP in teleosts and may also provide an important insight into the understanding of common regulatory mechanisms of feeding by AgRP among various animal species.

High concentration of E2 in breeding season facilitates *agrp1* expression

Our results suggest that *agrp1* and *npyb* show higher expressions under the breeding condition (LD) (Figure 3), but the experiments using juvenile female medaka (Figure 4) showed that expression levels of these two genes do not change according to the day-length itself but to the LD-induced sexual maturity. In addition, the present results indicate that the ovarian estrogen E2 upregulates *agrp1* expression mainly via the estrogen receptors *esr2a* that are co-expressed in some population of *agrp1* neurons in the hypothalamic nucleus NVT (Figure 5). Since LD female medaka (breeding) shows high blood concentration of E2 (Ikegami et al., 2022), this pathway may be important for breeding season-dependent feeding behavior. Especially, in teleost, main egg protein for nutrition is vitellogenin, whose expression is also facilitated by E2 (Tohyama et al., 2017). Taken together, it is suggested that E2 may synchronously regulate amount of food intake and female-specific reproductive signals (vitellogenin production and oocyte maturation), which plays a key role in reproductive success in oviparous animals.

In mammals, previous studies have reported on inconsistent effects of ovary and E2 on feeding. Ablation of ovary caused suppression of food intake in mice (Yu et al., 2020), whereas it caused no change in rats (Roesch, 2006). On the other hand, administration of E2 decreased food intake in both mice (Yu et al., 2020) and rats (Roesch, 2006). It should be noted that these laboratory rodents only exhibit short estrous cyclicity and have lost breeding seasonality, and the blood E2 concentration drastically changes in a few days (Nilsson et al., 2015). Thus, it is possible that the control mechanisms of feeding may be different between animals with short estrous cyclicity and those with breeding seasonality.

Furthermore, our present study suggests different control mechanisms of feeding between the animals with breeding seasonally and those without. Here, we showed that E2 directly modulates *agrp1* expression via *esr2a* receptors co-expressed in the *agrp1* neurons (**Figure 5C**), while in mice, *AgRP/NPY* neurons are reported to be suppressed by E2 indirectly via *esr1* (*era*)-expressing *Kiss1* neurons located in the hypothalamic arcuate nucleus (Qiu et al., 2018; Dubois et al., 2016; Yang et al., 2017). On the other hand, in medaka, expressions of all kinds of estrogen receptors are reported to be localized in NVT (Zempo et al., 2013), in which *agrp1* expression is also localized (**Figure 2B**). In addition, *esr2a* has been reported to be involved in the feedback regulation of follicle stimulating hormone in the pituitary and in the development of oviduct, and *esr2a* knockout females are completely infertile (Kayo et al., 2019). Our hypothesis that estrogen signaling via *esr2a* affects *agrp1* expression may highlight another important function of *esr2a* for reproduction, while a possibility still remains that *esr1*- and *esr2b*-expressing neurons also affect *agrp1* expression indirectly.

AgRP1 changes feeding behavior depending on LD-induced sexual maturity, which causes the increase in food intake in breeding season

Since our results thus far indicate the importance of *agrp1*, which shows upregulated expression directly stimulated by the ovarian E2 in breeding season, we examined phenotypes of *agrp1* knockout (*agrp1*^{-/-}) medaka. We found that *agrp1*^{-/-} medaka under the condition of breeding season eat less (**Figure 6B**) and spawn a smaller number of eggs (**Figure 6D**) than WT. Furthermore, the *agrp1*^{-/-} females did not show significant difference of food intake in LD and SD (**Figure 6C**). These results strengthen our hypothesis that *agrp1* is involved in the increased food intake in breeding season. Furthermore, *agrp1*^{-/-} female displayed light body weight (**Figure 6A**), accompanied by smaller ovary (**Figure 6E**) and low level of expression of gonadotropins, *fshb* and *lhb* (**Figure 6F**), which are considered to have caused smaller number of spawned eggs. All of these results support our hypothesis that AgRP1 plays an important role in breeding season-dependent feeding behavior, which culminates in normal reproduction.

In summary, by using a seasonal breeder medaka, we found evidence to suggest that long day-length facilitates ovarian maturation and E2 release, which upregulates *agrp1* expression of hypothalamic neurons to activate neural circuitry for feeding behavior and boost oocyte maturation. We propose that this kind of positive feedback control may be important for animals that spawn many eggs every day in the breeding season (**Figure 7**); medaka needs plenty of food for production of many eggs. In other words, the metabolic costs of producing eggs on a daily basis in medaka would inevitably require increased food intake. Indeed, previous study showed a need for high food intake for reproduction (Hasebe et al., 2016). It is expected that future studies will elucidate whether or not the present findings in medaka are applicable to other seasonal breeders as well.

Material and methods

Animals

Female and male wild-type d-rR medaka (*Oryzias latipes*) and *agrp1* knockout (*agrp1*^{-/-}) medaka were maintained in pairs or shoals at 27°C. Fish were fed three times a day with brine shrimp and flake food (Otohime B-2; San-u Fish Farm, Osaka, Japan). Their reproductive status was controlled by day-length (long day (LD; 14-hour light/10-hour dark): reproductive, short day (SD; 10-hour light/14-hour dark): non-reproductive). The light-on time was 8:00 AM. We used juvenile medaka (~5 weeks after fertilization) and adult medaka (>3 months after fertilization). Female medaka, which spawned at least three consecutive days, were used as reproductive ones. For the analysis of the effect of food restriction, reproductive and non-reproductive female medaka were fasted for

14 days or 10 days. Note that all medaka survived after food restriction. For the analyses of food intake, we food-restricted medaka for 6 hours after 10-min feeding in the morning and sampled their whole brains for subsequent experiments. Food-restricted medaka were sampled at the same time as the other fed medaka. All experiments and fish maintenance were conducted in accordance with the Guidelines for Proper Conduct of Animal Experiments (Science Council of Japan) and the protocols approved by the Animal Care and Use Committee of Graduate School of Science, the University of Tokyo (Permission number, 17-1, 20-6) and the Animal Care and Use Committee of Graduate School of Agriculture, Tokyo University of Agriculture and Technology (Permission number, R05-15, R06-27).

Food intake assay

Each 6-hour food-restricted medaka was put into a white cup with 100 mL breeding water and was habituated for 5 min. Then, we fed medaka by application of 200 μ L aliquots of food water containing brine shrimp in all-you-can-eat style and serve another aliquot once done with it, which is repeated N times (like the Japanese “Wanko soba”; so-called Japanese “Wanko soba” method). 10 min after the start, we stopped feeding medaka, placed a magnetic bar to stir the breeding water so that the shrimp concentration will be constant. Then we collected 10 mL aliquot from the experimental cup by using a micro pipette and transferred it to a conical tube. The conical tube was frozen overnight, and the leftover brine shrimp sunk in the bottom were counted by “shrimp-counter”. We counted the number of brine shrimp in the 200 μ L solution three times before and after the experiments, and the average number was used. The food intake was calculated as follows.

(Food Intake)

= (The average number of brine shrimp in the solution) * (number of aliquots, N)

- (number of leftover brine shrimp sunk in the bottom) * 10

Food intake was normalized by the average of LD or WT medaka.

“Shrimp-counter” system

The number of shrimps in the solution was counted using OpenCV3 library (Intel, Santa Clara, CA) run under a python script (Figure 1-Source code 1). This script was run under Anaconda 4.4.0 for windows running python 3.5.

RNA-sequencing

We collected two whole brains of LD or SD female in one tube (Note that pituitary was confirmed not to be included), and extracted total RNA by using NucleoSpin® RNA Plus kit (MACHEREY-NAGEL, Düren, Germany). cDNA was obtained by KAPA Stranded mRNA-Seq Kit (Nippon Genetics Co., Ltd, Tokyo, Japan) and KAPA Library preparation kit (Nippon Genetics Co., Ltd). Then, it was applied to a next generation sequencer Hiseq 2500 (Illumina, San Diego, CA), following the standard protocol of Illumina system. We selected the candidate genes judging from transcripts per million (TPM) for expression value in the obtained data using CLC workbench. We made volcano plot using R (R core team 2023) and Rstudio (2023) and colored dots, which indicate P-value < 0.05 and $|\log FC| > 1$. In addition, we made a heatmap of genes related to neuroendocrine system using DESeq2 (Love et al., 2014 [DOI](#)).

Histological analysis of the distribution of *agrp*- and *npy*-neurons in the brain

To analyze the distribution of *agrp*- and *npy*-expressing neurons, we performed *in situ* hybridization (ISH) for *agrp* and *npy* on frozen sections of reproductive medaka. In brief, female medaka was anesthetized (FA100, Bussan Animal Health Co., Ltd., Osaka, Japan), and its brain was picked up and fixed with 4% paraformaldehyde (PFA)/PBS. In analyses on *agrp2* expression, we performed perfusion-fixation by using 4% PFA/PBS. After incubation with 30% sucrose/PBS, brains were embedded in 5% low melting agar/ 20% sucrose/ PBS and sectioned at a thickness of 25 μ m. The sections were hybridized with *agrp1* (ENSORLG00000000398, 177 bases), *agrp2* (ENSORLG00000029106, 303 bases), *npya* (ENSORLG00000004649, 288 bases) and *npyb* (ENSORLG00000007880, 288 bases) specific digoxigenin (DIG)-labeled RNA probes and performed nitro blue tetrazolium (NBT)/ 5-bromo-4-chloro-3-indolyl-phosphate color-reaction (BCIP) after wash and incubation with anti-digoxigenin antibody (catalog. no. 11093274910; Roche; RRID: AB_514497) as previously reported (Zempo et al., 2013 [DOI](#)). Photographs were taken with a digital camera (DFC310FX; Leica Microsystems, Wetzlar, Germany) attached to an upright microscope (DM5000B; Leica Microsystems).

Histological analysis of *agrp1*-, *npy*- and estrogen receptor (*esr*)-expressing neurons

To examine whether *agrp1*-expressing neurons co-express *npya* and *esr*, we prepared *agrp1* fluorescein-labeled RNA probe and carried out double ISH as previously reported (Umatani et al., 2022 [DOI](#)). *esr* DIG-labeled probes were kindly given by Dr. Kayo (Kyoto Univ.), and we used *npya* DIG-labeled probe described in the previous paragraph. In brief, we made brain sections as described above and applied both *agrp1* fluorescein-labeled and each DIG-labeled RNA probes. Signals for *npya*, *esr1*, *esr2a* and *esr2b* were visualized by incubation with anti-digoxigenin antibody (catalog. no. 11207733910; Roche; RRID: AB_514500) and TSA Plus Cy3 System (TSA-Plus Cyanine 3 system, catalog. no. NEL744001KT, Akoya Biosciences, Marlborough, MA). After inactivation of Cy3 system by 3% H₂O₂, we applied peroxidase-conjugated anti-fluorescein antibody (catalog. no. 11426346910, Roche; RRID: AB_840257) on sections and performed TSA Plus biotin system (catalog. no. NEL749A001KT, Akoya Biosciences). Then, signals for *agrp1* were visualized by Alexa 488 conjugated streptavidin (catalog. no. S11223, Invitrogen). For counter-staining of cell nuclei, DAPI in PBS was applied on section. On the other hand, to examine whether *agrp2*-expressing neurons co-express *npya*, we used *npya* fluorescein-labeled RNA probe and *agrp2* DIG-labeled one. Double ISH of them was performed according to the same method described above. Fluorescent images were acquired with a confocal laser-scanning microscope (AXR, Nikon, Tokyo, Japan) using excitation and emission wavelengths of 405 nm and 429-474 nm for DAPI, 488 nm and 512-526 nm for Alexa 488, and 561 nm and 571-625 nm for Cy3, respectively. These were photographed at Tokyo University of Agriculture and Technology for Smart Core facility Promotion Organization.

Quantitative Real-time Polymerase Chain Reaction (RT-qPCR)

A Whole brain or a pituitary was collected from each medaka and total RNA was extracted by using FastGeneTM RNA basic kit (Nippon Genetics Co., Ltd) according to the manufacturer's instructions. For the juvenile medaka, we checked their sex as previously reported, and two samples of the same sex were mixed and used as one sample. Total RNA samples were reverse transcribed by FastGeneTM cDNA synthesis 5 $\times$ ReadyMix OdT according to the manufacturer's instructions. For the analyses of the brain, 1 μ L of cDNA diluted with 10-fold MQ was mixed with KAPA SYBR Fast qPCR kit and amplified with Lightcycler96 [Roche; 95L 150s (95L 10 sec, 60L 10 sec, 72L 15 sec) $\times$ 45 cycles]. For the analysis of the pituitary, 1 μ L of cDNA diluted with 5-fold MQ

was mixed with KAPA SYBR Fast qPCR kit and amplified with Lightcycler96 [Roche; 95L 150s (95L 10 sec, 60L 10 sec, 72L 10 sec) × 45 cycles]. The data was normalized by housekeeping gene, ribosomal protein s13 (*rps13*). Primer sequences were as follows:

- AgRP1 RT-PCR F1 CCAATTTCCAGTCACCGAAG
- AgRP1 RT-PCR R1 CTGGGTCCAACACAGAATCA
- AgRP2 RT-PCR F1 TTGTTGTGCTTCTTGCTGCT
- AgRP2 RT-PCR R1 ACAGAGCTCCAAACGGTGTC
- NPYa SE CTCATCACAAGACAGAGGTATGGG
- NPYa AS GGGTTGTAACCTGACTGTGGAAGTG
- NPYb SE CTGCCTGCTCCTCTGTTTTTCTC
- NPYb AS CACAGTGTCTGGGTTGTCTCTCTTC
- qPCR FSHb Fw new TGGAGATCTACAGGCGTCCGTAC
- qPCR FSHb Fw new AGCTCTCCACAGGGATGCTG
- qPCR LHb Fw new AGGGTATGTGACTGACGGATCCAC
- qPCR LHb Rv new TGCCTTACCAAGGACCCCTTGATG
- RPS13 SE GTGTTCCCACTTGGCTCAAGC
- RPS13 AS CACCAATTTGAGAGGGAGTGAGAC

Sham operations, ovariectomy, and E2 administrations

Ovariectomy and E2 administration were performed according to a previous study (Kayo *et al.*, 2020 [DOI](#)). Briefly, reproductive female medaka were anesthetized with 0.02% MS-222 (Sigma-Aldrich, St. Louis, MO) and their ovaries were excised via intraperitoneal operation. Sham operation group was anesthetized, received an abdominal incision without removing the ovaries, and received skin suture by using a silk thread. After checking that all Sham females spawn, we prepared three tanks; two tanks contained 7-8 OVX medaka, and one tank contained Sham medaka in 2 L breeding water in it. We dissolved β -Estradiol 1.4 mg in 1 mL EtOH (E2 stock) and dispensed 2 μ L of E2 stock or the same amount of 100% Ethanol for the control tank. Ethanol or E2 containing water were changed every day. After the steroid treatment for 5 days, the medaka were anesthetized, and their whole brains were collected for RT-qPCR analysis.

Generation of *agrp1* KO medaka lines

We generated *agrp1* KO medaka lines by using CRISPR/Cas9. Cas9 mRNA and tracer RNA were purchased from Integrated DNA Technologies (IDT, Coralville, IA). The guide RNA sequence for digestion by CRISPR/Cas9 complex was “CCTCACCAGCAGTCCTGCCTGG”.

Mixture of Cas9 protein, tracer RNA, CRISPR RNA, GFP mRNA diluted with PBS and 0.02% phenol red (final concentration: Cas9 protein; 500 ng/ μ L, tracer RNA; 100Lng/ μ L, CRISPR RNA; 50Lng/ μ L, GFP mRNA; 5Lng/ μ L) was injected into the cytoplasm of one- or two-cell-stage embryos (F0). To obtain homozygous transgenic offspring, the carriers were crossed with each other.

Measurement of body size and ovary size

We took photographs of fish bodies from the lateral side by using a digital camera MC120HD (Leica) and calculated the abdominal and body length by ImageJ. For the analysis of body length, we measured the length from the mouth to the base of the tail. Gonadosomatic index (GSI) was calculated as ovary weight/ body weight*100.

Immunohistochemistry using AgRP1 antibody

We made brain sections of WT and *agrp1* KO as described in “*Histological analysis of the distribution of agrp- and npy-neurons in the brain*”. After washing with PBST two times for 10 mins, sections were incubated with AgRP1 antibody (1:1000; Rabbit polyclonal AgRP (83-132) amide

(Human), Cat# H-003-53; RRID: AB_2313908, Phoenix Pharmaceutical, Burlingame, CA) /10% goat serum/PBS overnight. In the next day, slides were washed with PBST and incubated with anti-rabbit biotinylated goat antibody (1:200; Cat. #BA-1000; Vector Laboratories, Burlingame, CA) for 1 hour. Then we applied Alexa 488 conjugated streptavidin (1:500, Cat. #S11223, Invitrogen) and DAPI (1:2000; Dojin, Kumamoto, Japan). Photographs were taken with a digital camera (Leica Microsystems) attached to an upright microscope (Leica Microsystems).

Statistics

For the statistical analysis, we used Kyplot 5.0 software (Kyence, Tokyo, Japan) or R software (R core team 2023) with R studio (version 2023.06.0+421). For the comparison of the TPM, we used Student's t-test. For the comparison of *agrp1* and *npyb* expressions in OVX and E2 administrated medaka, Steel-Dwass test was used for multiple comparison of the expression level. In the other experiments, we used Mann-Whitney *U* test. In all statistical analysis, significance levels were described as follows: * $p < 0.05$, ** $p < 0.01$, and *** $p < 0.001$.

Data availability

All data supporting the findings of our present study and its supplementary information are available in the article or from the corresponding author upon reasonable request.

Code availability

The code of “Shrimp-counter” system is available in the present study, Figure 1-Source code 1 (see Methods).

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Additional information

Author contributions

Conceptualization: Y.T. and C.U.; Investigation: Y.T., S.T., H.W., S.Ki., S.Ka., and C.U.; Methodology: Y.T., H.W., T.K., S.Ka., and C.U.; Resources: Y.O., Y.H., and C.U.; Writing—original draft: Y.T. and C.U.; Writing—review and editing: Y.T., S.Ka., Y.O., Y.H., and C.U.; Project administration: C.U.; Supervision: S.Ka., Y.O., Y.H., and C.U.; and Funding acquisition: Y.T., S.Ka., Y.O., Y.H., and C.U.

Additional files

Supplementary files. Figure 1-Source code 1: The code of “Shrimp-counter” system **Figure 1-Supplementary Figure 1 “Wanko-soba” method for calculation of the amount of food intake of fish** (A) One medaka is put into a white cup with 100 mL breeding water and is habituated for 5 min. (B) Medaka is fed by application of 200 μ L aliquots of food water containing brine shrimp in all-you-can-eat style and serve another aliquot once done with it, which is repeated N times (like the Japanese “Wanko soba”). (C) Stop feeding 10 min after the start, take 10 mL out of the breeding water and transfer it to a conical tube. The conical tube is frozen overnight and the leftover brine shrimp sunk in the bottom are counted by “shrimp-counter”. The food intake is calculated as follows (also see Material and Methods). (Food Intake) = (The average number of brine shrimp in the solution) * (number of aliquots, N) - (number of leftover brine shrimp sunk in the bottom) * 10 Food intake was normalized by the average of LD medaka. **Figure 1-Supplementary Figure 2: Whole-brain gene expression in LD (n = 3) and SD (n = 3) female medaka.** (A) Volcano plot comparing whole-brain gene expression in LD (n = 3) and SD (n = 3) female medaka. Genes with higher expression in LD than in SD (P-value < 0.05 and logFC < -1) are shown in orange, and those with higher expression in SD than in LD (P-value < 0.05 and logFC > 1) are shown in blue. (B) Heatmap of representative genes, which are mainly related to neuroendocrine functions and show different expression patterns between LD and SD. (C and D) TPM value of *agrp1* and *npyb*. **Figure 2-Supplementary Figure 1 The number of npya-expressing neurons in the hypothalamus of fed and food-restricted (FR) medaka** (A) Total number of npya-expressing neurons in the hypothalamus of female medaka under normal fed condition (Fed; white; n = 4) or 2-weeks food restriction (FR; gray; n = 4). p = 0.02857, U = 16. (B) The number of npya-expressing neurons in the NVT of female medaka under normal fed condition (Fed; white; n = 4) or 2-weeks food restriction (FR; gray; n = 4). p = 0.1143, U = 14. (C) The number of npya-expressing neurons in the NRL of female medaka under normal fed condition (Fed; white; n = 4) or 2-weeks food restriction (FR; gray; n = 4). p = 0.02857, U = 16. (D) The number of npya-expressing neurons in the NAT of female medaka under normal fed condition (Fed; white; n = 4) or 2-weeks food restriction (FR; gray; n = 4). p = 0.02857, U = 16. Mann-Whitney U test, *p < 0.05. (E) *agrp2* expression of female medaka under normal fed condition (Fed) or 10 days food restriction (FR). Arrowheads indicate cell bodies of *agrp2*-expressing neurons. Scale bar: 100 μ m. **Figure 3-Supplementary Figure 1: Time course of the number of neurons showing ISH signals for *agrp1*.** (A) Time course of the number of neurons showing ISH signals for *agrp1* at each 30-min time point from the beginning of the chromogenic reaction in LD (orange; n = 5) and SD (blue; n = 5). The positive neurons were counted from 0 min to 300 min after application of NBT/BCIP (see Material and Methods), and the reaction saturated at 300 min onward. (B) Ratio of cell number at 90 min divided by that at 300 min. The relative cell number was not significantly different between LD and SD. (p = 0.2948, U = 18) **Figure 4-Supplementary Figure 1: Food intake of juvenile female does not show significant change according to the day-length.** (A) Method for counting food intake of juvenile. A fish was placed in a white cup containing 50 mL breeding water. After 5 mins habituation, 200 μ L shrimp water was applied. Fish was picked up 10 mins later by a dropper, and its fin was cut for the genotype of sex. The food intake was calculated by subtraction from the number of applied shrimps to that of the leftover shrimps. (B) Food intake of juvenile female medaka in LD (n = 9) and SD (n = 10) conditions. Each food intake was normalized to the average food intake of LD fish. (p = 0.7197, U = 50) Mann-Whitney U test. n.s., not significant. **Figure 5-Supplementary Figure 1: Food intake of OVX female does not show significant change according to the day-length.** Food intake of OVX female medaka in LD (n = 7) or SD (n = 6) conditions. Food intake was normalized to the average food intake of LD fish. (p = 0.7308, U = 18) Mann-Whitney U test. n.s., not significant. **Figure 6-Supplementary Figure 1: Mutation site of *agrp1* knockout medaka** (A) The DNA sequence of WT (*agrp1*^{+/+}) and mutated (*agrp1*^{-/-}) exon3 of *agrp1* gene. The nucleotide indicated with magenta represents an insertion. (B) The predicted amino acid sequence of *agrp1*^{+/+} and *agrp1*^{-/-}. Mutated amino acids are indicated with magenta. (C) AgRP1 immunoreactive signals were found in WT female brain but not in *agrp1*^{-/-} one. Upper row: WT, lower row: *agrp1*^{-/-}. Green: AgRP1 immunoreactive signals, and Blue: nuclei (DAPI). Scale bar: 50 μ m. **Figure 6-Supplementary Figure 2: Food intake of WT, *agrp1* hetero and homo knockout medaka** Food intake

of WT ($agrp1^{+/+}$, $n = 8$), hetero ($agrp1^{+/-}$, $n = 7$) and homo knockout ($agrp1^{-/-}$, $n = 5$) female medaka. $agrp1^{+/+}$ vs $agrp1^{-/-}$: $p = 0.5470$, $agrp1^{+/+}$ vs $agrp1^{+/-}$: $p = 0.009353$, $agrp1^{+/-}$ vs $agrp1^{-/-}$: $p = 0.01234$. The upper, middle, and lower bars show the third quartile, median, and the first quartile, respectively. Steel-Dwass test, * $p < 0.05$, ** $p < 0.01$. n.s., not significant. [↗](#)

References

- Adachi S., Yamada S., Takatsu Y., Matsui H., Kinoshita M., Takase K., Sugiura H., Ohtaki T., Matsumoto H., Uenoyama Y., Tsukamura H., Inoue K., Maeda K.-I (2007) **Involvement of Anteroventral Periventricular Metastin/Kisspeptin Neurons in Estrogen Positive Feedback Action on Luteinizing Hormone Release in Female Rats** *Journal of Reproduction and Development* **53**:367–378
- Adam C. L., Archer Z. A., Findlay P. A., Thomas L., Marie M (2002) **Hypothalamic gene expression in sheep for cocaine- and amphetamine-regulated transcript, pro-opiomelanocortin, neuropeptide Y, agouti-related peptide and leptin receptor and responses to negative energy balance** *Neuroendocrinology* **75**:250–256
- Amirjani S., Asemi Z., Bazarganipour F., Aramesh S., Allan H., Sayadi M., Tabatabaei M. S., Mohamadian Z., Zabti F., Iranpak N., Heydarzadeh A., Taghavi S. A., Badehnoosh B., Khashavi Z (2019) **Dietary intake and lifestyle behaviour in different phenotypes of polycystic ovarian syndrome: a case-control study** *J Hum Nutr Diet* **32**:413–421
- Andermann M. L., Lowell B. B (2017) **Toward a Wiring Diagram Understanding of Appetite Control** *Neuron* **95**:757–778
- Aponte Y., Atasoy D., Sternson S. M (2011) **AGRP neurons are sufficient to orchestrate feeding behavior rapidly and without training** *Nature Neuroscience* **14**:351–355
- Baldock P. A., Lee N. J., Driessler F., Lin S., Allison S., Stehrer B., Lin E. J., Zhang L., Enriquez R. F., Wong I. P., McDonald M. M., During M., Pierroz D. D., Slack K., Shi Y. C., Yulyaningsih E., Aljanova A., Little D. G., Ferrari S. L., Sainsbury A., Eisman J. A., Herzog H (2009) **Neuropeptide Y knockout mice reveal a central role of NPY in the coordination of bone mass to body weight** *PLoS One* **4**:e8415
- Blanco A. M., Soengas J. L (2021) **Leptin signalling in teleost fish with emphasis in food intake regulation** *Mol Cell Endocrinol* **526**:111209
- Broberger C., De Lecea L., Sutcliffe J. G., Hökfelt T. (1998) **Hypocretin/orexin- and melanin-concentrating hormone-expressing cells form distinct populations in the rodent lateral hypothalamus: relationship to the neuropeptide Y and agouti gene-related protein systems** *J Comp Neurol* **402**:460–474
- Chen P., Li B., Ou-Yang L (2022) **Role of estrogen receptors in health and disease** *Front Endocrinol (Lausanne)* **13**:839005
- Chen S. E., McMurtry J. P., Walzem R. L (2006) **Overfeeding-Induced Ovarian Dysfunction in Broiler Breeder Hens Is Associated with Lipotoxicity** *Poultry Science* **85**:70–81
- Chisada S., Kurokawa T., Murashita K., Rønnestad I., Taniguchi Y., Toyoda A., Sakaki Y., Takeda S., Yoshiura Y (2014) **Leptin receptor-deficient (knockout) medaka, *Oryzias latipes*, show chronic up-regulated levels of orexigenic neuropeptides, elevated food intake and stage specific effects on growth and fat allocation** *Gen Comp Endocrinol* **195**:9–20

- Clark J. T., Kalra P. S., Crowley W. R., Kalra S. P (1984) **Neuropeptide Y and human pancreatic polypeptide stimulate feeding behavior in rats** *Endocrinology* **115**:427–429
- Clarkson J., Herbison A. E (2009) **Oestrogen, kisspeptin, GPR54 and the pre-ovulatory luteinising hormone surge** *J Neuroendocrinol* **21**:305–311
- Conde-Sieira M., Soengas J. L (2016) **Nutrient Sensing Systems in Fish: Impact on Food Intake Regulation and Energy Homeostasis** *Frontiers in Neuroscience* **10**:603
- Dubois S. L., Wolfe A., Radovick S., Boehm U., Levine J. E (2016) **Estradiol Restrains Prepubertal Gonadotropin Secretion in Female Mice via Activation of ERα in Kisspeptin Neurons** *Endocrinology* **157**:1546–1554
- Egami N (1954) **Effect of artificial photoperiodicity on time of oviposition in the fish *Oryzias latipes***. *Annot Zool Japonenses* **27**:57–62
- Evans J. J., Anderson G. M (2012) **Balancing ovulation and anovulation: integration of the reproductive and energy balance axes by neuropeptides** *Human Reproduction Update* **18**:313–332
- Funahashi H., Takenoya F., Guan J. L., Kageyama H., Yada T., Shioda S (2003) **Hypothalamic neuronal networks and feeding-related peptides involved in the regulation of feeding** *Anat Sci Int* **78**:123–138
- Glenn Stanley B., Kyrkouli S. E., Lampert S., Leibowitz S. F. (1986) **Neuropeptide Y chronically injected into the hypothalamus: A powerful neurochemical inducer of hyperphagia and obesity** *Peptides* **7**:1189–1192
- Graham M., Shutter J. R., Sarmiento U., Sarosi I., Stark K. L (1997) **Overexpression of Agrt leads to obesity in transgenic mice** *Nat Genet* **17**:273–274
- Gray T. S., Morley J. E (1986) **Neuropeptide Y: Anatomical distribution and possible function in mammalian nervous system** *Life Sciences* **38**:389–401
- Hagan M. M., Rushing P. A., Pritchard L. M., Schwartz M. W., Strack A. M., Van Der Ploeg L. H., Woods S. C., Seeley R. J. (2000) **Long-term orexigenic effects of AgRP-(83---132) involve mechanisms other than melanocortin receptor blockade** *Am J Physiol Regul Integr Comp Physiol* **279**:R47–52
- Hahn T. M., Breininger J. F., Baskin D. G., Schwartz M. W (1998) **Coexpression of Agrp and NPY in fasting-activated hypothalamic neurons** *Nat Neurosci* **1**:271–272
- Hasebe M., Kanda S., Oka Y (2016) **Female-Specific Glucose Sensitivity of GnRH1 Neurons Leads to Sexually Dimorphic Inhibition of Reproduction in Medaka** *Endocrinology* **157**:4318–4329
- Ikegami K., Kajihara S., Umatani C., Nakajo M., Kanda S., Oka Y (2022) **Estrogen upregulates the firing activity of hypothalamic gonadotropin-releasing hormone (GnRH1) neurons in the evening in female medaka** *Journal of Neuroendocrinology* **34**:e13101
- Ilhnytska O., Argyropoulos G (2008) **The Role of the Agouti-Related Protein in Energy Balance Regulation** *Cellular and Molecular Life Sciences* **65**:2721

- Jennings K. J., de Lecea L. (2020) **Neural and Hormonal Control of Sexual Behavior** *Endocrinology* **161**
- Jeong I., Kim E., Kim S., Kim H.-K., Lee D.-W., Seong J. Y., Park H.-C (2018) **mRNA expression and metabolic regulation of npy and agrp1/2 in the zebrafish brain** *Neuroscience Letters* **668**:73–79
- Kalra S. P., Kalra P. S (1996) **Nutritional Infertility: The Role of the Interconnected Hypothalamic Neuropeptide Y–Galanin–Opioid Network** *Frontiers in Neuroendocrinology* **17**:371–401
- Kamijo M., Kojima K., Maruyama K., Konno N., Motohashi E., Ikegami T., Uchiyama M., Shioda S., Ando H., Matsuda K (2011) **Neuropeptide Y in tiger puffer (Takifugu rubripes): distribution, cloning, characterization, and mRNA expression responses to prandial condition** *Zoolog Sci* **28**:882–890
- Kanda S., Akazome Y., Matsunaga T., Yamamoto N., Yamada S., Tsukamura H., Maeda K.-i., Oka Y. (2008) **Identification of KiSS-1 Product Kisspeptin and Steroid-Sensitive Sexually Dimorphic Kisspeptin Neurons in Medaka (Oryzias latipes)** *Endocrinology* **149**:2467–2476
- Kanda S., Okubo K., Oka Y (2011) **Differential regulation of the luteinizing hormone genes in teleosts and tetrapods due to their distinct genomic environments--insights into gonadotropin beta subunit evolution** *Gen Comp Endocrinol* **173**:253–258
- Kauffman A. S., Rissman E. F (2004) **A critical role for the evolutionarily conserved gonadotropin-releasing hormone II: Mediation of energy status and female sexual behavior** *Endocrinology* **145**:3639–3646
- Kawai K., Fujita H., Sanchez G., Furusawa S., Umino T (2020) **Estimating the spawning season of black sea bream *Acanthopagrus schlegelii* in Hiroshima Bay, Japan, from temporal variation in egg density** *Fisheries Science* **86**:645–653
- Kayo D., Oka Y., Kanda S (2020) **Examination of methods for manipulating serum 17 β -Estradiol (E2) levels by analysis of blood E2 concentration in medaka (*Oryzias latipes*)** *Gen Comp Endocrinol* **285**:113272
- Kayo D., Zempo B., Tomihara S., Oka Y., Kanda S (2019) **Gene knockout analysis reveals essentiality of estrogen receptor β 1 (Esr2a) for female reproduction in medaka** *Sci Rep* **9**:8868
- Kelly M. J., Qiu J (2010) **Estrogen signaling in hypothalamic circuits controlling reproduction** *Brain Res* **1364**:44–52
- Kinoshita M., Murata K., Naruse K., Tanaka M (2009) **Medaka: biology, management, and experimental protocols** John Wiley & Sons
- Kirkwood R., Cumming D., Aherne F (1987) **Nutrition and puberty in the female** *Proceedings of the Nutrition Society* **46**:177–192
- Kohn D., Yada T (2012) **Arcuate NPY neurons sense and integrate peripheral metabolic signals to control feeding** *Neuropeptides* **46**:315–319
- Krashes M. J., Koda S., Ye C., Rogan S. C., Adams A. C., Cusher D. S., Maratos-Flier E., Roth B. L., Lowell B. B (2011) **Rapid, reversible activation of AgRP neurons drives feeding behavior in**

mice Journal of Clinical Investigation **121**:1424–1428

Larsson T. A., Tay B. H., Sundström G., Fredriksson R., Brenner S., Larhammar D., Venkatesh B (2009) **Neuropeptide Y-family peptides and receptors in the elephant shark, *Callorhinchus milii* confirm gene duplications before the gnathostome radiation** *Genomics* **93**:254–260

Liu R., Kinoshita M., Adolphi M. C., Scharl M (2019) **Analysis of the Role of the Mc4r System in Development, Growth, and Puberty of Medaka** *Frontiers in Endocrinology* **10**:213

Love M. I., Huber W., Anders S (2014) **Moderated estimation of fold change and dispersion for RNA-seq data with DESeq2** *Genome Biology* **15**:550

Lynn S. E., Stampis T. B., Barrington W. T., Weida N., Hudak C. A (2010) **Food, stress, and reproduction: short-term fasting alters endocrine physiology and reproductive behavior in the zebra finch** *Horm Behav* **58**:214–222

Marks J. L., Li M., Schwartz M., Porte D., Baskin D. G (1992) **Effect of fasting on regional levels of neuropeptide Y mRNA and insulin receptors in the rat hypothalamus: An autoradiographic study** *Molecular and Cellular Neuroscience* **3**:199–205

Marks J. L., Waite K (1997) **Intracerebroventricular neuropeptide Y acutely influences glucose metabolism and insulin sensitivity in the rat** *J Neuroendocrinol* **9**:99–103

Melo A. C., Ramsdell J. S (2001) **Sexual dimorphism of brain aromatase activity in medaka: induction of a female phenotype by estradiol** *Environmental Health Perspectives* **109**:257–264

Merry B., Holehan A. M (1979) **Onset of puberty and duration of fertility in rats fed a restricted diet** *Reproduction* **57**:253–259

Micevych P. E., Meisel R. L (2017) **Integrating Neural Circuits Controlling Female Sexual Behavior** *Front Syst Neurosci* **11**:42

Mitani Y., Kanda S., Akazome Y., Zempo B., Oka Y (2010) **Hypothalamic Kiss1 but not Kiss2 neurons are involved in estrogen feedback in medaka (*Oryzias latipes*)** *Endocrinology* **151**:1751–1759

Muroi Y., Ishii T (2016) **A novel neuropeptide Y neuronal pathway linking energy state and reproductive behavior** *Neuropeptides* **59**:1–8

Naftolin F., Garcia-Segura L. M., Horvath T. L., Zarnovszky A., Demir N., Fadiel A., Leranath C., Vondracek-Klepper S., Lewis C., Chang A., Parducz A (2007) **Estrogen-induced hypothalamic synaptic plasticity and pituitary sensitization in the control of the estrogen-induced gonadotrophin surge** *Reprod Sci* **14**:101–116

Nakajima K., Cui Z., Li C., Meister J., Cui Y., Fu O., Smith A. S., Jain S., Lowell B. B., Krashes M. J., Wess J (2016) **Gs-coupled GPCR signalling in AgRP neurons triggers sustained increase in food intake** *Nat Commun* **7**:10268

Nilsson M. E., Vandenput L., Tivesten Å., Norlén A. K., Lagerquist M. K., Windahl S. H., Börjesson A. E., Farman H. H., Poutanen M., Benrick A., Maliqueo M., Stener-Victorin E., Ryberg H., Ohlsson C (2015) **Measurement of a Comprehensive Sex Steroid Profile in Rodent Serum by High-Sensitive Gas Chromatography-Tandem Mass Spectrometry** *Endocrinology* **156**:2492–2502

- Ollmann M. M., Wilson B. D., Yang Y. K., Kerns J. A., Chen Y., Gantz I., Barsh G. S (1997) **Antagonism of central melanocortin receptors in vitro and in vivo by agouti-related protein** *Science* **278**:135–138
- Opazo R., Plaza-Parrochia F., Cardoso Dos Santos G. R., Carneiro G. R. A., Sardela V. F., Romero J., Valladares L. (2018) **Fasting Upregulates npy, agrp, and ghsl Without Increasing Ghrelin Levels in Zebrafish (Danio rerio) Larvae** *Front Physiol* **9**:1901
- Peute J., van der Gaag M. A., Lambert J. G. D. (1978) **Ultrastructure and lipid content of the liver of the zebrafish, Brachydanio rerio, related to vitellogenin synthesis** *Cell and Tissue Research* **186**:297–308
- Qian S., Chen H., Weingarh D., Trumbauer M. E., Novi D. E., Guan X., Yu H., Shen Z., Feng Y., Frazier E., Chen A., Camacho R. E., Shearman L. P., Gopal-Truter S., MacNeil D. J., Van der Ploeg L. H., Marsh D. J. (2002) **Neither agouti-related protein nor neuropeptide Y is critically required for the regulation of energy homeostasis in mice** *Mol Cell Biol* **22**:5027–5035
- Qiu J., Rivera H. M., Bosch M. A., Padilla S. L., Stincic T. L., Palmiter R. D., Kelly M. J., Rønnekleiv O. K (2018) **Estrogenic-dependent glutamatergic neurotransmission from kisspeptin neurons governs feeding circuits in females** *eLife* **7**
- Robinson E. J., Rugh R (1943) **The reproductive processes of the fish, Oryzias latipes** *The Biological Bulletin* **84**:115–125
- Roesch D. M (2006) **Effects of selective estrogen receptor agonists on food intake and body weight gain in rats** *Physiol Behav* **87**:39–44
- Rønnestad I., Gomes A. S., Murashita K., Angotzi R., Jonsson E., Volkoff H (2017) **Appetite-Controlling Endocrine Systems in Teleosts** *Frontiers in Endocrinology* **8**:73
- Schwartz M. W., Woods S. C., Porte D., Seeley R. J., Baskin D. G (2000) **Central nervous system control of food intake** *Nature* **404**:661–671
- Shainer I., Buchshtab A., Hawkins T. A., Wilson S. W., Cone R. D., Gothilf Y (2017) **Novel hypophysiotropic AgRP2 neurons and pineal cells revealed by BAC transgenesis in zebrafish** *Scientific Reports* **7**:44777
- Shainer I., Michel M., Marquart G. D., Bhandiwad A. A., Zmora N., Ben-Moshe Livne Z., Zohar Y., Hazak A., Mazon Y., Forster D., Hollander-Cohen L., Cone R. D., Burgess H. A., Gothilf Y (2019) **Agouti-Related Protein 2 Is a New Player in the Teleost Stress Response System** *Current Biology* **29**:2009–2019
- Shiozaki K., Kawabe M., Karasuyama K., Kurachi T., Hayashi A., Ataka K., Iwai H., Takeno H., Hayasaka O., Kotani T., Komatsu M., Inui A (2020) **Neuropeptide Y deficiency induces anxiety-like behaviours in zebrafish (Danio rerio)** *Sci Rep* **10**:5913
- Shutter J. R., Graham M., Kinsey A. C., Scully S., Lüthy R., Stark K. L (1997) **Hypothalamic expression of ART, a novel gene related to agouti, is up-regulated in obese and diabetic mutant mice** *Genes Dev* **11**:593–602
- Söderberg C., Wraith A., Ringvall M., Yan Y. L., Postlethwait J. H., Brodin L., Larhammar D (2000) **Zebrafish genes for neuropeptide Y and peptide YY reveal origin by chromosome duplication from an ancestral gene linked to the homeobox cluster** *J Neurochem* **75**:908–918

- Song Y., Cone R. D (2007) **Creation of a genetic model of obesity in a teleost** *FASEB Journal* **21**:2042–2049
- Song Y., Golling G., Thacker T. L., Cone R. D (2003) **Agouti-related protein (AGRP) is conserved and regulated by metabolic state in the zebrafish, Danio rerio** *Endocrine* **22**:257–265
- Stone D. B., Cherry M. J., Martin J. A., Cohen B. S., Miller K. V (2017) **Breeding chronology and social interactions affect ungulate foraging behavior at a concentrated food resource** *PLoS One* **12**:e0178477
- Sundstrom G., Larsson T. A., Brenner S., Venkatesh B., Larhammar D (2008) **Evolution of the neuropeptide Y family: new genes by chromosome duplications in early vertebrates and in teleost fishes** *General and Comparative Endocrinology* **155**:705–716
- Takahashi K. A., Cone R. D (2005) **Fasting induces a large, leptin-dependent increase in the intrinsic action potential frequency of orexigenic arcuate nucleus neuropeptide Y/Agouti-related protein neurons** *Endocrinology* **146**:1043–1047
- Temple J. L., Rissman E. F (2000) **Brief refeeding restores reproductive readiness in food-restricted female musk shrews (Suncus murinus)** *Horm Behav* **38**:21–28
- Tohyama S., Miyagawa S., Lange A., Ogino Y., Mizutani T., Ihara M., Tanaka H., Tatarazako N., Kobayashi T., Tyler C. R., Iguchi T (2016) **Evolution of estrogen receptors in ray-finned fish and their comparative responses to estrogenic substances** *J Steroid Biochem Mol Biol* **158**:189–197
- Tohyama S., Ogino Y., Lange A., Myosho T., Kobayashi T., Hirano Y., Yamada G., Sato T., Tatarazako N., Tyler C. R., Iguchi T., Miyagawa S (2017) **Establishment of estrogen receptor 1 (ESR1)-knockout medaka: ESR1 is dispensable for sexual development and reproduction in medaka, Oryzias latipes** *Development Growth & Differentiation* **59**:552–561
- Tsuyuki A. T.U. (2018) **Assessment of ichthyofauna at oyster rafts in Hiroshima Bay, Japan, using underwater video cameras** *Aquaculture Science* **66**:267–274
- Umatani C., Yoshida N., Yamamoto E., Akazome Y., Mori Y., Kanda S., Okubo K., Oka Y (2022) **Co-existing Neuropeptide FF and Gonadotropin-Releasing Hormone 3 Coordinately Modulate Male Sexual Behavior** *Endocrinology* **163**
- Volk K. M., Pogrebna V. V., Roberts J. A., Zachry J. E., Blythe S. N., Toporikova N (2017) **High-Fat High-Sugar Diet Disrupts the Preovulatory Hormone Surge and Induces Cystic Ovaries in Cycling Female Rats.** *J Endocr Soc* **1**:1488–1505
- Wintermantel T. M., Campbell R. E., Porteous R., Bock D., Gröne H. J., Todman M. G., Korach K. S., Greiner E., Pérez C. A., Schütz G., Herbison A. E (2006) **Definition of estrogen receptor pathway critical for estrogen positive feedback to gonadotropin-releasing hormone neurons and fertility** *Neuron* **52**:271–280
- Yang J. A., Stires H., Belden W. J., Roepke T. A (2017) **The Arcuate Estrogen-Regulated Transcriptome: Estrogen Response Element-Dependent and -Independent Signaling of ERα in Female Mice** *Endocrinology* **158**:612–626
- Yokobori E., Azuma M., Nishiguchi R., Kang K. S., Kamijo M., Uchiyama M., Matsuda K (2012) **Neuropeptide Y stimulates food intake in the Zebrafish, Danio rerio** *Journal of*

Neuroendocrinology **24**:766–773

Yu K., He Y., Hyseni I., Pei Z., Yang Y., Xu P., Cai X., Liu H., Qu N., Liu H., He Y., Yu M., Liang C., Yang T., Wang J., Gourdy P., Arnal J. F., Lenfant F., Xu Y., Wang C (2020) **17 β -estradiol promotes acute refeeding in hungry mice via membrane-initiated ER α signaling** *Mol Metab* **42**:101053

Zempo B., Kanda S., Okubo K., Akazome Y., Oka Y (2013) **Anatomical distribution of sex steroid hormone receptors in the brain of female medaka** *Journal of Comparative Neurology* **521**:1760–1780

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Reviewer #1 (Public review):

Summary:

The authors use the teleost medaka as an animal model to study the effect of seasonal changes in day-length on feeding behaviour and oocyte production. They report a careful analysis how day-length affects female medakas and a thorough molecular genetic analysis of genes potentially involved in this process. They show a detailed analysis of two genes and include a mutant analysis of one gene to support their conclusions

Strengths:

The authors pick their animal model well and exploit the possibilities to examine in this laboratory model the effect of a key environmental influence, namely the seasonal changes of day-length. The phenotypic changes are carefully analysed and well controlled. The mutational analysis of the *agrp1* by a ko-mutant provides important evidence to support the conclusions. Thus this report exceeds previous findings on the function of *agrp1* and *npyb* as regulators of food-intake and shows how in medaka these genes are involved in regulating the organismal response to an environmental change. It thus furthers our understanding on how animals react to key exogenous stimuli for adaptation.

Weaknesses:

The authors are too modest when it comes to underscoring the importance of their findings. Previous animal models used to study the effect of these neuropeptides on feeding behaviour have either lost or were most likely never sensitive to seasonal changes of day-length. Considering the key importance of this parameter on many aspects of plant and animal life it could be better emphasised that a suitable animal model is at hand that permits this. The molecular characterization of the *agrp1* ko-mutant that the authors have generated lacks some details that would help to appreciate the validity of the mutant phenotype. Additional data would help in this respect.

Comments on revisions:

The authors dealt adequately with the comments and suggestions of this reviewer.

<https://doi.org/10.7554/eLife.100996.2.sa3>

Reviewer #2 (Public review):

Summary:

The authors investigated the mechanisms behind breeding season-dependent feeding behavior using medaka, a well-known photoperiodic species, as a model. Through a combination of molecular, cellular, and behavioral analyses, including tests with mutants,

they concluded that AgRP1 plays a central role in feeding behavior, mediated by ovarian estrogenic signals.

Strengths:

This study offers valuable insights into the neuroendocrine mechanisms that govern breeding season-dependent feeding behavior in medaka. The multidisciplinary approach, which includes molecular and physiological analyses, enhances the scientific contribution of the research.

Comments on revised version:

My concerns from the first review have been addressed. The manuscript's key points are clearly presented, and the conclusions are readily comprehensible

<https://doi.org/10.7554/eLife.100996.2.sa2>

Reviewer #3 (Public review):

Summary:

Understanding the mechanisms whereby animals restrict the timing of their reproduction according to day length is a critical challenge given that many of the most relevant species for agriculture are strongly photoperiodic. However, the principal animal models capable of detailed genetic analysis do not respond to photoperiod so this has inevitably limited progress in this field. The fish model medaka occupies a uniquely powerful position since its reproduction is strictly restricted to long days and it also offers a wide range of genetic tools for exploring, in depth, various molecular and cellular control mechanisms.

For these reasons, this manuscript by Tagui and colleagues is particularly valuable. It uses the medaka to explore links bridging photoperiod, feeding behaviour and reproduction. The authors demonstrate that in female, but not male medaka, photoperiod-induced reproduction is associated with an increase in feeding, presumably explained by the high metabolic cost of producing eggs on a daily basis during the reproductive period. Using RNAseq analysis of the brain, they reveal that the expression of the neuropeptides agrp and npy that have been previously implicated in the regulation of feeding behaviour in mice, are upregulated in the medaka brain during exposure to long photoperiod conditions. Unlike the situation in mouse, these two neuropeptides are not coexpressed in medaka neurons and food deprivation in medaka led to increases in agrp but also a decrease in npy expression. Furthermore, the situation in fish may be more complicated than in mouse due to the presence of multiple gene paralogs for each neuropeptide. Exposure to long day conditions increases agrp1 expression in medaka as the result of increases in the number of neurons expressing this neuropeptide, while the increase in npyb levels results from increased levels of expression in the same population of cells. Using ovariectomized medaka and in situ hybridization assays, the authors reveal that the regulation of agrp1 involves estrogen acting via the estrogen receptor esr2a. Finally, a loss of agrp1 function mutant is generated where the female mutants fail to show the characteristic increase in feeding associated with long day enhanced reproduction as well as yielding reduced numbers of eggs during spawning.

Strengths:

This manuscript provides important foundational work for future investigations aiming to elucidate the coordination of photoperiod sensing, feeding activity and reproduction function. The authors have used a combination of approaches with a genetic model that is particularly well suited to studying photoperiodic dependent physiology and behaviour. The data are clear and the results are convincing and support the main conclusions drawn. The

findings are relevant not only for understanding photoperiodic responses but also provide more general insight into links between reproduction and feeding behaviour control.

The manuscript has been further strengthened by the inclusion of additional information according to my advice: The analysis of ovariectomized female fish and juvenile fish has now been reported in terms of their feeding behaviour and so provide a complete view of the position of this feeding regulatory mechanism in the context of reproduction status. Furthermore, the discussion section has been expanded to speculate on the functional significance of linking feeding behaviour control with reproductive function. Modifications made in order to address technical concerns of the other 2 reviewers have also significantly strengthened the presentation of this work.

Weaknesses:

These have now been addressed in the revised version.

<https://doi.org/10.7554/eLife.100996.2.sa1>

Author response:

The following is the authors' response to the original reviews

Public Reviews:

Reviewer #1 (Public review):

Summary:

The authors use the teleost medaka as an animal model to study the effect of seasonal changes in day-length on feeding behaviour and oocyte production. They report a careful analysis of how day-length affects female medakas and a thorough molecular genetic analysis of genes potentially involved in this process. They show a detailed analysis of two genes and include a mutant analysis of one gene to support their conclusions

Strengths:

The authors pick their animal model well and exploit the possibilities to examine in this laboratory model the effect of a key environmental influence, namely the seasonal changes of day-length. The phenotypic changes are carefully analysed and well-controlled. The mutational analysis of the agrp1 by a ko-mutant provides important evidence to support the conclusions. Thus this report exceeds previous findings on the function of agrp1 and npyb as regulators of food-intake and shows how in medaka these genes are involved in regulating the organismal response to an environmental change. It thus furthers our understanding of how animals react to key exogenous stimuli for adaptation.

Weaknesses:

The authors are too modest when it comes to underscoring the importance of their findings. Previous animal models used to study the effect of these neuropeptides on feeding behaviour have either lost or were most likely never sensitive to seasonal changes of day length. Considering the key importance of this parameter on many aspects of plant and animal life it could be better emphasised that a suitable animal model is at hand that permits this. The molecular characterization of the agrp1 ko-mutant that the authors have generated lacks some details that would help to appreciate the validity of the mutant phenotype. Additional data would help in this respect.

We would like to thank Reviewer #1 for the really constructive advice. In the revised manuscript, we provided more information on the molecular characterization of the *agrp1* KO-mutant and to emphasize the importance of our present animal model that permits the analysis of neuropeptide effects on feeding behavior in response to seasonal changes of day length.

Reviewer #2 (Public review):

Summary:

The authors investigated the mechanisms behind breeding season-dependent feeding behavior using medaka, a well-known photoperiodic species, as a model. Through a combination of molecular, cellular, and behavioral analyses, including tests with mutants, they concluded that AgRP1 plays a central role in feeding behavior, mediated by ovarian estrogenic signals.

Strengths:

This study offers valuable insights into the neuroendocrine mechanisms that govern breeding season-dependent feeding behavior in medaka. The multidisciplinary approach, which includes molecular and physiological analyses, enhances the scientific contribution of the research.

Weaknesses:

While medaka is an appropriate model for studying seasonal breeding, the results presented are insufficient to fully support the authors' conclusions.

Specifically, methods and data analyses are incomplete in justifying the primary claims:

- the procedure for the food intake assay is unclear;*
- the sample size is very small;*
- the statistical analysis is not always adequate.*

Additionally, the discussion fails to consider the possible role of other hormones that may be involved in the feeding mechanism.

We would like to thank Reviewer #2 for the helpful comments. As the reviewer suggested, we revised the paragraph describing the procedure for the food intake assay to make it much easier for the readers to understand in the revised manuscript. In Figure 1-Supplementary figure 2, RNAseq was performed to search for the candidate neuropeptides, and that's why the sample size was the minimum. On the other hand, each group in the other experiments consist of $n \geq 5$ samples, which is usually accepted to be adequate sample size in various studies (cf. Kanda et al., Gen Comp Endocrinol., 2011, Spicer et al., Biol Reprod., 2017). As for the statistical analyses, we revised our manuscript so that the readers may be convinced with the validity of our statistical analyses.

Reviewer #3 (Public review):

Summary:

Understanding the mechanisms whereby animals restrict the timing of their reproduction according to day length is a critical challenge given that many of the most relevant species for agriculture are strongly photoperiodic. However, the principal animal models capable of detailed genetic analysis do not respond to photoperiod so this has inevitably limited progress in this field. The fish model medaka occupies a

uniquely powerful position since its reproduction is strictly restricted to long days and it also offers a wide range of genetic tools for exploring, in depth, various molecular and cellular control mechanisms.

For these reasons, this manuscript by Tagui and colleagues is particularly valuable. It uses the medaka to explore links bridging photoperiod, feeding behaviour, and reproduction. The authors demonstrate that in female, but not male medaka, photoperiod-induced reproduction is associated with an increase in feeding, presumably explained by the high metabolic cost of producing eggs on a daily basis during the reproductive period. Using RNAseq analysis of the brain, they reveal that the expression of the neuropeptides agrp and npy that have been previously implicated in the regulation of feeding behaviour in mice are upregulated in the medaka brain during exposure to long photoperiod conditions. Unlike the situation in mice, these two neuropeptides are not co-expressed in medaka neurons, and food deprivation in medaka led to increases in agrp but also a decrease in npy expression. Furthermore, the situation in fish may be more complicated than in mice due to the presence of multiple gene paralogs for each neuropeptide. Exposure to long-day conditions increases agrp1 expression in medaka as the result of increases in the number of neurons expressing this neuropeptide, while the increase in npyb levels results from increased levels of expression in the same population of cells. Using ovariectomized medaka and in situ hybridization assays, the authors reveal that the regulation of agrp1 involves estrogen acting via the estrogen receptor esr2a. Finally, a loss of agrp1 function mutant is generated where the female mutants fail to show the characteristic increase in feeding associated with long-day enhanced reproduction as well as yielding reduced numbers of eggs during spawning.

Strengths:

This manuscript provides important foundational work for future investigations aiming to elucidate the coordination of photoperiod sensing, feeding activity, and reproduction function. The authors have used a combination of approaches with a genetic model that is particularly well suited to studying photoperiodic-dependent physiology and behaviour. The data are clear and the results are convincing and support the main conclusions drawn. The findings are relevant not only for understanding photoperiodic responses but also provide more general insight into links between reproduction and feeding behaviour control.

Weaknesses:

Some experimental models used in this study, namely ovariectomized female fish and juvenile fish have not been analysed in terms of their feeding behaviour and so do not give a complete view of the position of this feeding regulatory mechanism in the context of reproduction status. Furthermore, the scope of the discussion section should be expanded to speculate on the functional significance of linking feeding behaviour control with reproductive function.

We would like to thank Reviewer #3 for the insightful advice. We added several pertinent sentences describing the ovariectomized female fish and juvenile fish, and our revised manuscript will give more complete view of their feeding regulatory mechanism in the context of reproduction status. In addition, we revised the discussion section to incorporate the valuable suggestion of the Reviewer #3.

Recommendations for the authors:

Reviewer #1 (Recommendations for the authors):

General: the text could profit from a careful editing of errors, including adjusting singular and plural status of nouns and verbs: examples are line 107 noun, line 96 verb suitable text editing software is available to do this task

Thank you for your suggestion. We thoroughly read the entire manuscript and corrected such errors in the revised manuscript.

*As medaka is a unique genetic vertebrate model to study seasonal effects, it would be interesting to know whether the authors found novel or rather unexpected genes with a differential expression between LD and SD. It is understandable that the authors focused on *argrp1* and *npyb*, as these have already been well studied in mammalian models although not in this context. Novel insights with genes previously not implicated in feeding regulation could underscore the unique nature of medaka as a model.*

We appreciate your kind comments, which we found really encouraging to us. Since we focused on feeding-related peptides, we did not find any novel genes that have not been reported.

*ISH is unreliable as a methodology to quantify expression levels. Yet the authors use this to compare fed and starved females to compare expression levels of *argrp1*. They use a temporal staining comparison and compare 90-minute and 300-minute staining reactions. However, they do not explain why they use the 90-minute staining time point and why 300 minutes of staining is the "saturation point of staining". They should provide compelling data for their claim and the selection of time points or else refrain from using these (at best) semi-quantitative ISH and provide more detailed (using serial sections) data to quantify the number of expressing cells.*

Anyhow, the quantification of mRNA expression levels may not be that significant when trying to compare different states of gene function, as translational and post-translational steps can have large effects on gene function. This should be discussed adequately.

Thank you very much for your comments. We conducted ISH by using medaka under LD or SD, not using those under fed or starved conditions. In addition, our previous study demonstrated that the slopes of the increase in the number of cells stained by ISH are also different if there is a difference in the expression level (Mitani et al., 2010). Although we do not have quantitative data of cell numbers, we confirmed that the number of cells expressing *argrp1* was saturated around 300 mins in our preliminary experiments, and therefore we terminated the chemogenic reactions at 300 mins. Based on these, we compared the cell ratio of 90 min (beginning of coloring) /300 min (saturation). However, since this analysis may not be worth discussing in detail, we moved this part to the supplementary figure as the reviewer suggested.

*The molecular characterization of the *argrp1* ko mutant is a bit thin.*

*Line 221: "We obtained *argrp1*^{-/-} medaka, which has lots of amino acid changes in functional site for AgRP1" is a bit vague as a description for the ko-mutation. It would be really helpful if the authors could provide a scheme showing the wt protein with the relevant functional sites alongside the presumptive mutant protein.*

How did the authors verify the molecular nature of their mutation? They should use suitable antibodies and western-blot analysis (maybe reagents from Shainer et al., 2019 work in medaka); in case this is not possible they could isolate & clone the mutant

transcript and use in-vitro translation systems to show that the presumptive mutant protein can actually be translated from this transcript. Another strategy could be to use a second non-allelic and (hopefully) non-complementing mutation (ko1/ko2 heterozygotes for example) to show that ko-mutation acts the way the authors presume. The authors mention agrp1 ko medaka lines (plural!) in line 520, thus they may have an additional ko allele at hand.

Thank you very much for your comments. We explained the mutation site in Figure 6-Supplementary Figure 1 (A: DNA sequences and B: predicted amino acid sequence, of WT and mutants). In addition, we added immunohistochemistry data of WT and mutant using anti-AgRP antibody (Figure 6-Supplementary Figure 1C). While AgRP-immunoreactive signals were observed in WT, those were not in *agrp1*^{-/-}. This result suggests that AgRP1 is not functional in *agrp1*^{-/-}.

Presumably, the authors analysed heterozygous agrp1^{+/-} females and found they are as wt. If so the authors should say so.

Yes, we analyzed food intake of *agrp1*^{+/-}. We added a supplementary figure (Figure 6-Supplementary Figure 2) and a sentence in L. 233-234.

How about agrp1^{-/-} medaka males: do they show a discernible phenotype?

We analyzed the phenotypes of *agrp1*^{-/-} males but did not describe the results, since the present paper only focused on female-specific feeding behavior.

agrp1^{-/-} females show no significant sensitivity of food intake to day length (Figure 6C). Does their (reduced) oocyte production react to day length? With other words: how much of the seasonal sensitivity is left in agrp1^{-/-} females. The authors suggest that E2 acts upstream of agrp1 and therefore some seasonality may still be left in agrp1^{-/-} females.

Although *agrp1*^{-/-} female is suggested to display abnormal seasonality of food intake, *agrp1*^{-/-} female in LD spawns and that in SD does not, indicating that seasonality of gonadal maturation still remains in *agrp1*^{-/-} female.

The authors show that fshb and lhb are downregulated in agrp1^{-/-} females. Is this also the case in wt females at SD?

Thank you very much for your comment. As described above, *agrp1*^{-/-} can spawn, which indicates that mechanisms for the downregulation of gonadotropins in *agrp1*^{-/-} may be different from that in SD female.

Figure 1_Supplementary Figure 2: the trends are visible in B and C, however, there is quite some variance between LD1, 2, and 3; the same for SD 1, 2, and 3. Can the authors give an explanation for this?

Since the data for LD1, 2, and 3 (SD1, 2, and 3) were obtained from different individual fish, the variance may be reasonable. We conducted expression analyses by using RNA-seq to find candidate genes that show larger differences than individual ones.

Figure 7E: the ovaries are difficult to see and the size bar in the wt picture is missing.

Thank you very much for your comments. We added a scale bar in the wt picture.

509 ff: the authors do not describe what exactly the "sham operation" encompasses: were the females just anesthetised or was there an actual operation without removing the ovaries?

The sham operation group was anesthetized, received an abdominal incision without removing the ovaries, and received skin suture by using a silk thread. We added this explanation in the Method section.

*519 ff: was the *agrp1*^{-/-} ko induced in the d-rR strain to have the same genetic background as the wt fish?*

Exactly. As the reviewer pointed out, the genetic background of *agrp1*^{-/-} was the same as that of WT.

Minor points (Text edits):

Line 42: change "when" into "where".

Line: 54 "under the fixed appropriate ambient temperature" change into "while keeping an appropriate temperature constant".

Line 55: here it would be good to briefly explain what long-day and short-day is so that the reader has an idea about the changes required without having to scroll down to the M&M section. For example LD 14/10 light-dark cycle, SD 10/14 light-dark cycle.

Line 88: change "measurement" into "measuring".

Line 96 change eats -> eat.

Line 107 change female -> females.

We deeply appreciate the reviewer's suggestions described above. We corrected them as the reviewer suggested (L. 42, L. 54, L. 55, L. 89, L. 96, L. 107).

Line 144-145: the sentence "since hypothalamic npy control..." does not make sense. Please correct.

Thank you very much for your suggestion. We corrected the sentence so that it makes sense (L. 145-146).

Line 180 and 185: the term here should be "LD induced sexual activity" rather than maturity. Age is the main determinant of maturity whereas light (LD) determines activity, in other words SD females are sexually mature if they are post-puberty stage.

Thank you very much for your suggestion. Since the sentence "LD-induced sexual maturity" made the reviewer confused, we corrected the sentence "substance(s) from LD-induced mature ovary" or "ovarian maturity". Even though SD females are at post-puberty stage, their ovaries are immature and do not possess mature oocytes (L. 181).

*Line 222: the authors should include the relevant information about the females: presumably *agrp1*.*

In Line 226-228, we explained the phenotypes of *agrp1* knockout and added information for AgRP1 protein in Figure 6-Supplementary figure 1C.

Lines 449 ff: authors should state that the analysis was done in females, instead of just writing "medaka". This is also in line with the preceding paragraph of the M&M section.

Thank you very much for your suggestions. We corrected the sentence as the reviewer suggested (L.469)

Line 305: change like other mammals -> like in mammals.

Thank you very much for your suggestion. We corrected the sentence as the reviewer suggested (L. 320)

Reviewer #2 (Recommendations for the authors):

(1) The procedure of the food intake assay is not clear.

- Habituation Period: Medaka were placed into a white cup containing 100 mL of water and allowed to habituate for 5 minutes. However, is 5 minutes sufficient to reduce stress in the fish? A stressed fish does not exhibit the same feeding behavior as an unstressed one.

Thank you for your comment. We confirmed that 5 minutes is enough for habituation in medaka, since medaka can swim freely in a few minutes after replacement from the tank and show normal feeding behavior.

- Feeding Protocol: Medaka were fed with 200 μ L aliquots of brine shrimp-containing water. This procedure was repeated multiple times. How many times was this feeding procedure repeated? Was it 3, 10, or 100 times?

Although there was a small variation in each trial, we usually applied tubes about 5 times or so.

- Brine Shrimp Counting: You collected 10 mL of the breeding water to count the number of uneaten brine shrimp. Can you confirm that sampling 10% of the total volume is representative? Were any tests conducted to validate this? Given that you developed an automated tool to count the brine shrimp, why didn't you count them in all 100 mL?

The reason for collecting 10 mL is to collect the leftover shrimp as soon as possible. Ten mins after the start of the experiment, we quickly placed a magnetic bar to stir the breeding water so that the shrimp concentration will be constant. Then we collected 10 mL aliquot from the experimental cup by using a micro pipette. In preliminary trials, we applied shrimps, the amount of which is almost the same as that applied to WT medaka in LD, to a white cup containing 100 mL water, and we divided it into 10 mL and 90 mL aliquots and separately counted the number of shrimps in each aliquot. Here, we confirmed that the variance between the numbers calculated by counting the shrimps in 10 mL aliquot and the total volume of 100 mL falls within the range of the variance of total applied shrimp. Thus, our present counting method can be considered reasonable.

- Brine Shrimp Aliquot Measurement: You mentioned counting the number of brine shrimp in the 200 μ L solution three times before and after the experiments. What does this mean? Did you use this procedure to calculate the mean number of brine shrimp in each 200 μ L aliquot?

Thank you for your comment. As the reviewer commented, to calculate the mean number of brine shrimp in each 200 μ L aliquot, we counted the number of brine shrimp in the 200 μ L

solution three times before and after the experiments.

- How did you normalize the food intake data? This procedure is not detailed in the methods section.

Thank you very much for pointing it out. We normalized food intake by subtracting the amount of shrimp by the average of those in LD or WT fish. This explanation was added in the Method section (L. 439).

(2) Sample Size. Various tests were conducted with a low number of medaka (e.g., 2 brains for RNA-seq, 8 females for ovariectomy). Are these sample sizes sufficient to draw reliable conclusions?

In Figure 1-Supplementary figure 2, RNAseq was performed to search for the candidate neuropeptides, and that's why the sample size was the minimum; we pooled two brains as one sample and used three samples per group. On the other hand, each group in the other experiments consist of $n \geq 5$ samples, which is usually accepted to be adequate sample size in various studies (cf. Kanda et al., Gen Comp Endocrinol., 2011, Spicer et al., Biol Reprod., 2017).

(3) Statistical Analysis.

- The authors used both parametric and non-parametric tests but did not specify how they assessed the normal distribution of the data. For example, if I understood correctly, a t-test was used to compare a small dataset ($n=3$). In such cases, a U-test would be more appropriate.

Thank you for your comment. As for Figure 1 -Supplementary Figure 2C, we showed the graphs just to show you candidates. To avoid misunderstanding, we deleted statistical statements in that panel.

- It is unclear why the Steel-Dwass test was used instead of the Kruskal-Wallis test for comparing agrp1 and npyb expressions in control, OVX, and E2-administered medaka.

While the authors mentioned using non-parametric tests, they did not specify in which contexts or conditions they were applied.

Thank you very much for your comment. Kruskal-Wallis test statistically shows whether or not there are differences among any of three groups. To perform multiple comparisons among the three groups, we used Steel-Dwass test.

- The results section lacks details on the statistical tests used, including the specific test (e.g., Z, U, or W values) and degrees of freedom.

Thank you for your comment. As the reviewer pointed out, we added such statements in all the figure legends containing statistics.

(4) Previous studies have shown that photoperiod treatments alter the production of various hormones in medaka (e.g., Lucon-Xiccato et al., 2022; Shimmura et al., 2017), some of which, like growth hormone (GH), have been shown to influence feeding behavior (Canosa et al., 2007).

In your RNA-seq analysis, did you observe any changes in the expression of genes involved in other hormone synthesis pathways, such as pituitary hormones (GH and TSH), leptin, or ghrelin (e.g., see Volkoff, 2016; Blanco, 2020; Bertolucci et al., 2019)?

Including such evidence in the discussion would provide a broader perspective on the hormonal regulation of food intake in medaka.

We appreciate your constructive comments. Unfortunately, since we performed RNA-seq using the whole brain after removal of the pituitary, we could not check such changes in the expression of pituitary hormone-related genes. As additional information about the feeding-related hormones, leptin did not show significant difference in our RNA-seq analysis, and we could not analyze ghrelin because ghrelin has not been annotated in medaka (NCBI and ensembl).

Reviewer #3 (Recommendations for the authors):

There are some parts of the study that need to be developed further in order to provide a more comprehensive analysis.

(1) In the juvenile as well as ovariectomized female fish, the authors should confirm experimentally whether day length influences feeding activity.

Thank you very much for your suggestion. We analyzed feeding behavior of juvenile (Figure 4-Supplementary Figure 1) and OVX female (Figure 5-Supplementary Figure 1). As shown in these figures, food intake in juvenile and OVX were not significantly different between LD and SD.

(2) More discussion as to the relevance of increasing feeding activity to support reproductive functions such as sustained egg production would be valuable. One assumes the metabolic costs of producing eggs on a daily basis in this species would inevitably require increased food intake. Is this a reasonable prediction?

We deeply appreciate your suggestion. We strongly agree with this argument, and we added such discussion in “Discussion” section (L. 406-408).

Editor's note:

Should you choose to revise your manuscript, if you have not already done so, please include full statistical reporting including exact p-values wherever possible alongside the summary statistics (test statistic and df) and 95% confidence intervals. These should be reported for all key questions and not only when the p-value is less than 0.05 in the main manuscript.

We appreciate the editor's suggestion. We added P-value in the main manuscript, where statistical analyses were performed. In addition, we described test statics in the figure legends. We did not use df values for the statistics used in the present analyses, and therefore did not describe it in the main text.

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